LBM_CODE

A modular lattice Boltzmann solver on Kokkos

Scheme description, methodology and validation

Abstract

This document describes the numerical schemes, the software architecture and the complete validation record of `LBM_CODE`, a portable lattice Boltzmann solver written in C++20 on top of Kokkos. The code targets NVIDIA GPUs and CPU clusters from a single source. It supports five lattices, two streaming schemes (including the in-place Esoteric Pull), two population storage conventions, four collision operators for the fluid (BGK, TRT, raw multiple relaxation time and central moments), a passive-scalar module for heat transfer and a magnetohydrodynamic module built on a vector-valued distribution. Every scheme documented here is accompanied by the quantitative test that establishes it, and every stated limitation is one that has been measured rather than assumed.

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1 Scope and design

1.1 What the code is

`LBM_CODE` solves the weakly compressible Navier–Stokes equations by the lattice Boltzmann method, optionally coupled to a passive scalar (temperature) and to a magnetic field. It is written once and compiled for any Kokkos backend: Serial, Threads, OpenMP, CUDA, HIP or SYCL. Scalar precision is selected at configure time.

1.2 Governing equations

The fluid obeys

$$\partial_t \rho + \nabla\cdot(\rho \boldsymbol{u}) = 0, \tag{1}$$
$$\partial_t(\rho \boldsymbol{u}) + \nabla\cdot(\rho \boldsymbol{u}\boldsymbol{u}) = -\nabla p + \nabla\cdot\boldsymbol{\tau} + \boldsymbol{F}, \tag{2}$$

with $p = \rho c_s^2$ and $\boldsymbol{\tau}$ the viscous stress. When the thermal module is active a scalar T is transported by

$$\partial_t T + \nabla\cdot(\boldsymbol{u}T) = D\,\nabla^2 T, \tag{3}$$

and couples back to the flow through the Boussinesq body force. When the magnetohydrodynamic module is active the magnetic field $\boldsymbol{b}$ obeys the induction equation

$$\partial_t \boldsymbol{b} = \nabla\times(\boldsymbol{u}\times\boldsymbol{b}) + \eta\,\nabla^2\boldsymbol{b}, \qquad \nabla\cdot\boldsymbol{b} = 0, \tag{4}$$

and acts on the flow through the Lorentz force $\boldsymbol{j}\times\boldsymbol{b}$ with $\boldsymbol{j} = \nabla\times\boldsymbol{b}$.

1.3 Architectural rules

Seven rules govern the implementation. They are stated here because most of the design decisions in later sections follow from them rather than from the physics.

1. **No virtual functions in device code.** Lattice, streaming scheme, equilibrium, forcing, storage convention and collision operator are all compile-time policies that inline into a single kernel. Dynamic dispatch exists only in a host-side factory above the solver. A virtual call inside a Kokkos kernel runs, but it blocks inlining of the entire collision operator.

2. **Never pin the Kokkos memory layout.** Population fields are declared `View<Real**>` with extents `(node, q)` and the *default* layout. Kokkos then selects `LayoutLeft` on CUDA/HIP — node index fastest, hence a structure of arrays with coalesced access — and `LayoutRight` on CPU backends, giving an array of structures with good cache locality per node. One declaration is correct on both; hardcoding either costs performance on the other.

3. **Streaming and collision are one fused kernel.** The in-place streaming schemes cannot have a separate streaming pass, so nothing in the solver is written in a way that would assume one.

4. **Parity is a compile-time parameter.** Esoteric Pull addresses different storage slots on even and odd time steps. That decision must fold away at compile time, so `FluidSolver::step` dispatches to `run_step<0>` or `run_step<1>`.

5. **Periodicity is index wrapping, not halo copying.** See §3.3.
6. **Boundary conditions are alternative collision operators** applied to flagged cells, giving one branch per node rather than one per direction.
7. **The reference implementation is never deleted.** The two-lattice streaming scheme is retained permanently and cross-checked against Esoteric Pull after every step.

2 Lattices

2.1 Definition and constraints

A lattice is a set of Q velocities $\boldsymbol{c}_i$ with weights w_i and a lattice sound speed c_s. To recover the Navier–Stokes equations at second order the set must satisfy

$$\textstyle\sum_i w_i = 1, \qquad \sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}, \tag{5}$$

$$\textstyle\sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} = 0, \qquad \sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} c_{i\delta} = c_s^4 (\delta_{\alpha\beta}\delta_{\gamma\delta} + \delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma}). \tag{6}$$

Five lattices are provided. **D2Q5** and **D3Q7** satisfy (5) but *not* (6): they cannot support Navier–Stokes and are used only for scalar transport and for the magnetic field, where only the second moment is required.

lattice	Q	D	c_s^2	weights	Navier–Stokes
D2Q5	5	2	1/3	1/3; 1/6	no
D2Q9	9	2	1/3	4/9; 1/9; 1/36	yes
D3Q7	7	3	1/4	1/4; 1/8	no
D3Q27	27	3	1/3	8/27; 2/27; 1/54; 1/216	yes

Table 1: The four lattices. Note that D3Q7 has $c_s^2 = 1/4$, not 1/3.

2.2 Velocity sets

The complete velocity sets follow, in the Esoteric Pull ordering of §2.3. These tables are generated from the same data the code compiles against, and the identities in the captions are re-verified in exact rational arithmetic at generation time.

2.3 The direction-ordering contract

Esoteric Pull walks opposite pairs as `for (i = 1; i < Q; i += 2)`. This requires that the rest velocity occupies index 0 and that opposite directions occupy *adjacent* indices, so that

$$\mathrm{opp}(0) = 0, \qquad \mathrm{opp}(i) = \begin{cases} i+1 & i \text{ odd} \\ i-1 & i \text{ even} \end{cases} \tag{7}$$

is pure integer arithmetic, with no lookup table in the kernel. This is a *contract*: every moment matrix in the code is generated against it.

Reordering the directions of a lattice is a column permutation of the moment transform matrices together with the matching permutation of f_i and f_i^{eq}. Every moment, raw or central, is therefore invariant, and only the per-direction post-collision expressions are permuted. This was verified numerically in exact rational arithmetic before the orderings were adopted.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0)$	$\frac{1}{3}$	0	3	$(0,1)$	$\frac{1}{6}$	4
1	$(1,0)$	$\frac{1}{6}$	2	4	$(0,-1)$	$\frac{1}{6}$	3
2	$(-1,0)$	$\frac{1}{6}$	1				

Table 2: D2Q5: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{3}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0)$	$\frac{4}{9}$	0	5	$(1,1)$	$\frac{1}{36}$	6
1	$(1,0)$	$\frac{1}{9}$	2	6	$(-1,-1)$	$\frac{1}{36}$	5
2	$(-1,0)$	$\frac{1}{9}$	1	7	$(1,-1)$	$\frac{1}{36}$	8
3	$(0,1)$	$\frac{1}{9}$	4	8	$(-1,1)$	$\frac{1}{36}$	7
4	$(0,-1)$	$\frac{1}{9}$	3				

Table 3: D2Q9: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{3}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

2.4 Compile-time verification

Weights are stored as exact rationals ($w_i =$ `w_num`$[i]/$`w_den`) and c_s^2 as `cs2_num/cs2_den`, so all the identities above are checked at compile time in exact integer arithmetic rather than in floating point:

```
static_assert(check_pairing<L>(),       "opposite directions not adjacent");
static_assert(check_unique<L>(),        "duplicated velocity");
static_assert(check_w_sum<L>(),         "weights do not sum to 1");
static_assert(check_second_moment<L>(), "<cc> != cs2 I");
static_assert(check_third_moment<L>(),  "<ccc> != 0");
static_assert(check_fourth_moment<L>() == L::supports_navier_stokes, "");
```

The last line is an equality rather than a truth test: D2Q5 and D3Q7 are *required* to fail fourth-order isotropy, and a lattice whose flag disagrees with its actual moments will not compile.

3 Streaming

3.1 Two-lattice (reference)

Two arrays, $2Q$ reals per node. Semantics are *pull*: the kernel gathers the populations that have arrived at node n,

$$f_i^{\text{in}}(n) = f_i^{\text{src}}(n - \boldsymbol{c}_i), \tag{8}$$

collides, and writes locally to the destination array, which is then swapped. This scheme is obviously race-free and obviously correct. It is not the production scheme, but it is retained permanently: when an exotic collision operator produces incorrect results, being able to switch to a streaming scheme that cannot be wrong is what makes the fault bisectable.

3.2 Esoteric Pull

Esoteric Pull [4] stores a single copy of the populations and updates them in place. Write $A[n][s]$ for storage slot s at node n, and consider one opposite pair $(i, i+1)$ with i odd. The partner node is *always* $n + \boldsymbol{c}_i$, at both parities; only which of the two slots lives remotely alternates.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0,0)$	$\frac{1}{4}$	0	4	$(0,-1,0)$	$\frac{1}{8}$	3
1	$(1,0,0)$	$\frac{1}{8}$	2	5	$(0,0,1)$	$\frac{1}{8}$	6
2	$(-1,0,0)$	$\frac{1}{8}$	1	6	$(0,0,-1)$	$\frac{1}{8}$	5
3	$(0,1,0)$	$\frac{1}{8}$	4				

Table 4: D3Q7: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{4}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0,0)$	$\frac{8}{27}$	0	9	$(1,-1,0)$	$\frac{1}{54}$	10	18	$(0,-1,1)$	$\frac{1}{54}$	17
1	$(1,0,0)$	$\frac{2}{27}$	2	10	$(-1,1,0)$	$\frac{1}{54}$	9	19	$(1,1,1)$	$\frac{1}{216}$	20
2	$(-1,0,0)$	$\frac{2}{27}$	1	11	$(1,0,1)$	$\frac{1}{54}$	12	20	$(-1,-1,-1)$	$\frac{1}{216}$	19
3	$(0,1,0)$	$\frac{2}{27}$	4	12	$(-1,0,-1)$	$\frac{1}{54}$	11	21	$(1,1,-1)$	$\frac{1}{216}$	22
4	$(0,-1,0)$	$\frac{2}{27}$	3	13	$(1,0,-1)$	$\frac{1}{54}$	14	22	$(-1,-1,1)$	$\frac{1}{216}$	21
5	$(0,0,1)$	$\frac{2}{27}$	6	14	$(-1,0,1)$	$\frac{1}{54}$	13	23	$(1,-1,1)$	$\frac{1}{216}$	24
6	$(0,0,-1)$	$\frac{2}{27}$	5	15	$(0,1,1)$	$\frac{1}{54}$	16	24	$(-1,1,-1)$	$\frac{1}{216}$	23
7	$(1,1,0)$	$\frac{1}{54}$	8	16	$(0,-1,-1)$	$\frac{1}{54}$	15	25	$(-1,1,1)$	$\frac{1}{216}$	26
8	$(-1,-1,0)$	$\frac{1}{54}$	7	17	$(0,1,-1)$	$\frac{1}{54}$	18	26	$(1,-1,-1)$	$\frac{1}{216}$	25

Table 5: D3Q27: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{3}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

Consistency. At an even step node n writes its outgoing f_i into $A[n + \boldsymbol{c}_i][i{+}1]$; at the next (odd) step node $m = n + \boldsymbol{c}_i$ reads f_i from $A[m][i{+}1]$, which is exactly the odd-step rule. The scheme closes.

Race freedom. Every slot has exactly one reader and one writer, and they are the same node. No temporary buffer and no synchronisation within a step are required.

Implicit bounce-back. Halfway bounce-back sets $f_i^{\text{out}} = f_{i+1}^{\text{in}}$ and $f_{i+1}^{\text{out}} = f_i^{\text{in}}$. Substituting into Table 6, every write lands back in the slot it was read from: the entire update is the identity. *Solid cells are therefore skipped outright* — no load, no store, no arithmetic. The population a fluid node emits toward a wall sits untouched for one step and is read back as the opposite direction on the next, placing the wall exactly midway between the two cells.

Memory. Q reals per node against $2Q$ for two-lattice. At 512^3 with D3Q27 in FP64 this is the difference between 29 GB and 58 GB, i.e. between fitting and not fitting on a 40 GB device.

3.3 Periodicity by index wrapping

Esoteric Pull writes into its neighbours' storage. A halo-copy treatment of periodicity would therefore require a *two-way* exchange whose set of exchanged slots depends on the parity. The code instead resolves periodicity by wrapping the neighbour index, which removes that entire class of bug:

$$\text{shift}(v) = \begin{cases} v+n & v<h \\ v-n & v \ge h+n \\ v & \text{otherwise} \end{cases} \quad \text{(periodic)}, \qquad \text{clamp}(v,0,s-1) \quad \text{(otherwise)}. \tag{9}$$

even step t	odd step t
$f_i \leftarrow A[n][i]$	$f_i \leftarrow A[n][i{+}1]$
$f_{i+1} \leftarrow A[n+\boldsymbol{c}_i][i{+}1]$	$f_{i+1} \leftarrow A[n+\boldsymbol{c}_i][i]$
$A[n+\boldsymbol{c}_i][i{+}1] \leftarrow f_i^{\mathrm{out}}$	$A[n+\boldsymbol{c}_i][i] \leftarrow f_i^{\mathrm{out}}$
$A[n][i] \leftarrow f_{i+1}^{\mathrm{out}}$	$A[n][i{+}1] \leftarrow f_{i+1}^{\mathrm{out}}$

Table 6: The Esoteric Pull access pattern, one opposite pair. Each parity reads exactly two slots and writes those same two, crossed.

A halo layer is allocated only in non-periodic directions, where edge cells genuinely need somewhere out-of-domain to read from. A fast path skips the wrapping entirely for nodes away from every face.

4 Population storage

Two conventions are supported.

Raw the arrays hold f_i.

Shifted the arrays hold $g_i = f_i - w_i$.

The shift exists for conditioning. Populations are $O(w_i) \sim 10^{-1}$, while the collision operates on differences $O(10^{-6})$; forming $f_i - f_i^{\mathrm{eq}}$ in FP32 therefore discards most of the mantissa. The momentum sum is worse, since $\sum_i \boldsymbol{c}_i f_i$ cancels terms of order 10^{-1} down to 10^{-2}. Storing g_i removes both cancellations, because $\sum_i \boldsymbol{c}_i w_i = 0$ exactly and hence

$$\sum_i \boldsymbol{c}_i\, g_i \;\equiv\; \sum_i \boldsymbol{c}_i\, f_i \tag{10}$$

identically, while every summand is small. The density follows from $\rho = 1 + \sum_i g_i$.

Almost nothing else changes. Streaming is a pure copy and is untouched. Halfway bounce-back is $g_i \leftarrow g_{\mathrm{opp}(i)}$, untouched, because $w_i = w_{\mathrm{opp}(i)}$. The Guo source term is already an $O(F)$ quantity and is untouched. Only the equilibrium and the density reconstruction differ.

Measured effect. Poiseuille flow at the magic relaxation time, $H = 32$, in FP32:

storage	FP64	FP32
raw	1.433×10^{-13}	5.969×10^{-4}
shifted	1.626×10^{-14}	$\mathbf{2.778 \times 10^{-5}}$
gain	8.8×	**21.5×**

The validation suite asserts that gain rather than merely tolerating its absence: a regression to raw storage fails the build.

For the passive scalar the same argument applies but there is *no universal reference.* Density is always near unity; a temperature scale is whatever the problem says it is. The scalar collision therefore exposes T_{ref} as a runtime parameter, defaulting to 0, which reproduces unshifted storage exactly.

5 Equilibria

This section states, for every lattice, the equilibrium that the code actually evaluates. The distinction matters in one place in particular: the moment-space operators of §6.3 *never form equilibrium populations at all.* Their equilibrium is defined directly in moment space, and the corresponding populations exist only implicitly, as the inverse transform of those moments.

5.1 The ϕ formulation

Each populations-space equilibrium policy supplies only $\phi_i(\boldsymbol{u})$, defined by

$$f_i^{\mathrm{eq}} = w_i\,\rho\,(1+\phi_i), \tag{11}$$

from which both storage conventions of §4 follow once:

$$\begin{aligned} \text{raw:}\quad & f_i^{\mathrm{eq}} = w_i\,\rho\,(1+\phi_i), && (12)\\ \text{shifted:}\quad & f_i^{\mathrm{eq}} - w_i = w_i\big(\delta\rho + (1+\delta\rho)\,\phi_i\big), \qquad \delta\rho = \rho - 1. && (13)\end{aligned}$$

Form (13) is written that way deliberately: every term is small, so it never forms the difference of two $O(w_i)$ quantities. Writing it as $f_i^{\mathrm{eq}} - w_i$ would be algebraically identical and numerically useless.

5.2 Second-order (truncated Hermite) — D2Q9, D3Q27

The default for BGK, TRT and the MHD fluid operator, on all three Navier–Stokes lattices:

$$\boxed{f_i^{\mathrm{eq}} = w_i\,\rho\left[1 + \frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_i\cdot\boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u}\cdot\boldsymbol{u}}{2c_s^2}\right]} \tag{14}$$

With $c_s^2 = 1/3$ on all three, this is $f_i^{\mathrm{eq}} = w_i\rho\big[1 + 3(\boldsymbol{c}_i\cdot\boldsymbol{u}) + \frac{9}{2}(\boldsymbol{c}_i\cdot\boldsymbol{u})^2 - \frac{3}{2}\boldsymbol{u}^2\big]$. It is correct to $O(u^2)$; its central moments carry $O(u^3)$ defects, which is what the Galilean-invariance test of §10.4 measures and what the moment operators remove.

5.3 Product form — D2Q9 and D3Q27

The complete (fourth-order) equilibrium, equivalent to `index_equilibrium = 4` of the symbolic generator. Writing $a_\alpha = c_{i\alpha}^2 - c_s^2$,

$$\boxed{\begin{aligned} f_i^{\mathrm{eq}} = w_i\rho\Big[1 &+ \frac{c_{ix}u_x + c_{iy}u_y}{c_s^2} + \frac{a_x u_x^2 + a_y u_y^2 + 2c_{ix}c_{iy}u_xu_y}{2c_s^4}\\ &+ \frac{a_x c_{iy} u_x^2 u_y + a_y c_{ix} u_x u_y^2}{2c_s^6} + \frac{a_x a_y\, u_x^2 u_y^2}{4c_s^8}\Big]\end{aligned}} \tag{15}$$

Its central moments are *exactly* Maxwellian for all nine representable k_{pq} — verified symbolically, 0 of 9 deviating — so it is fully Galilean invariant on this lattice.

D3Q27: sixth order. D3Q27 is a product lattice, so the same construction runs to sixth order — the highest its 27 velocities admit. The symbolic generator writes it as a sum of Hermite tensors of orders one to six. That sum was checked, in exact rational arithmetic, against the factorised form

$$\boxed{f_i^{\mathrm{eq}} = \rho\,\psi(c_{ix},u_x)\,\psi(c_{iy},u_y)\,\psi(c_{iz},u_z), \qquad \begin{aligned}\psi(0,u) &= 1 - c_s^2 - u^2,\\ \psi(\pm1,u) &= \tfrac{1}{2}\big(c_s^2 + u^2 \pm u\big),\end{aligned}} \tag{16}$$

and the two are *identical, term for term.* The one-dimensional factor ψ is simply the three-point distribution carrying the exact moments $\{1,\, u,\, c_s^2 + u^2\}$. Equation (16) is what the code evaluates: three multiplications instead of six Hermite tensors, bit-for-bit the same polynomial. Dividing each factor by its one-dimensional weight $\{\frac{1}{6}, \frac{2}{3}, \frac{1}{6}\}$ puts it in the ϕ form of §5.1, since $w_i = w^{1\mathrm{D}}(c_{ix}) w^{1\mathrm{D}}(c_{iy}) w^{1\mathrm{D}}(c_{iz})$ on a product lattice:

$$\phi_i = \chi(c_{ix}, u_x)\,\chi(c_{iy}, u_y)\,\chi(c_{iz}, u_z) - 1, \qquad \chi(0, u) = 1 - \tfrac{3}{2}u^2, \quad \chi(\pm 1, u) = 1 + 3u(\pm 1 + u). \tag{17}$$

The defining property. The central moments of (15) and (16) are *exactly* Maxwellian in every representable moment:

$$k^{\mathrm{eq}}_{pqr} = \begin{cases} \rho\,(c_s^2)^{\,n}, & p, q, r \text{ all even, } n = \#\{p, q, r \text{ equal to } 2\}, \\ 0, & \text{otherwise,} \end{cases} \tag{18}$$

so 8 of the 27 D3Q27 moments are nonzero and the remaining 19 vanish identically. Every Galilean-invariance defect of the truncation (14) is absent.

Where these are used. (15) and (16) are collected as `HighOrderEquilibrium<L>`, which selects the highest order each lattice admits. This is the default for the MHD operators of §6.4. The BGK and TRT defaults remain (14), and the moment-space operators of §5.4 reach the same equilibrium without evaluating any of these forms explicitly.

5.4 Equilibria in moment space — D2Q9, D3Q27

The MRT and central-moment operators specify their equilibrium as *moments*, in closed form. Writing $\boldsymbol{u}_b$ for the basis velocity ($\mathbf{0}$ for raw moments, $\boldsymbol{u}$ for central) and $\delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}_b$, the equilibrium moment in the stored variable is a product of one-dimensional factors,

$$\boxed{E_{pqr} = \rho\, Q_p(\delta u_x)\, Q_q(\delta u_y)\, Q_r(\delta u_z) \;-\; [\text{shifted}]\; A_p(u_{bx})\, A_q(u_{by})\, A_r(u_{bz})} \tag{19}$$

with the triples depending on which basis the lattice uses:

lattice	basis	$Q_{0,1,2}$	$A_{0,1,2}$
D2Q9, D3Q27	product, $\varphi_2 = C^2 - c_s^2$	$1,\ \delta u,\ \delta u^2$	$1,\ -u_b,\ u_b^2$

Two consequences are worth stating explicitly.

Central moments on D2Q9 and D3Q27. Here $\delta\boldsymbol{u} = \mathbf{0}$, so $Q = (1, 0, 0)$ and (19) collapses to

$$E_{0\ldots 0} = \rho, \qquad E_{pqr} = 0 \;\text{ otherwise} \qquad \text{(raw storage).} \tag{20}$$

One nonzero equilibrium moment. This is the property that makes the operator cheap: no equilibrium populations, no stored K^{eq} vector, no $Q \times Q$ matrices. It was verified symbolically — 0 of 9 deviating on D2Q9, 0 of 27 on D3Q27 — and it is equivalent to relaxing toward (15) and its D3Q27 counterpart.

Raw MRT. Here $\boldsymbol{u}_b = \mathbf{0}$ and $\delta\boldsymbol{u} = \boldsymbol{u}$, so $Q = (1, u, u^2)$ on the product basis and $(1, u, c_s^2 + u^2)$ on the monomial one — the raw moments of a Maxwellian centred on $\boldsymbol{u}$. Note this gives raw MRT the *complete* equilibrium, not a truncated one, which is why it recovers most of the Galilean-invariance gain on its own (§10.4).

5.5 Scalar transport — D2Q5, D3Q7

The passive scalar carries only its zeroth moment, $T = \sum_i g_i$, and is advected by a velocity supplied from outside:

$$\boxed{g_i^{\mathrm{eq}} = w_i\, T\left(1 + \frac{\boldsymbol{c}_i\!\cdot\!\boldsymbol{u}}{c_s^2}\right)} \qquad D = c_s^2\Big(\frac{1}{\omega} - \frac{1}{2}\Big). \tag{21}$$

Explicitly, with $c_s^2 = 1/3$ on D2Q5 and $c_s^2 = 1/4$ on D3Q7,

$$\text{D2Q5:}\quad g_i^{\mathrm{eq}} = w_i T\big(1 + 3\,\boldsymbol{c}_i\!\cdot\!\boldsymbol{u}\big), \qquad w = \big(\tfrac{1}{3};\ \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}\big), \tag{22}$$

$$\text{D3Q7:}\quad g_i^{\mathrm{eq}} = w_i T\big(1 + 4\,\boldsymbol{c}_i\!\cdot\!\boldsymbol{u}\big), \qquad w = \big(\tfrac{1}{4};\ \tfrac{1}{8}\ \text{(six times)}\big). \tag{23}$$

In the shifted storage of §4 the code evaluates

$$h_i^{\mathrm{eq}} = w_i\left[\delta T + (T_{\mathrm{ref}} + \delta T)\,\frac{\boldsymbol{c}_i\!\cdot\!\boldsymbol{u}}{c_s^2}\right], \qquad \delta T = T - T_{\mathrm{ref}}, \tag{24}$$

which is (21) minus $w_i T_{\mathrm{ref}}$, written so that no term is the difference of two $O(w_i T)$ quantities.

Why first order. The truncation in (21) is deliberate. D2Q5 and D3Q7 have an isotropic second moment but not an isotropic fourth-order one, and on D3Q7 every velocity has a single nonzero component, so $(\boldsymbol{c}_i\!\cdot\!\boldsymbol{u})^2$ reduces to $c_\alpha^2 u_\alpha^2$ and the cross terms of the $\boldsymbol{uu}$ tensor cannot be represented at all. A second-order term would add an anisotropic error rather than accuracy. The cost is an $O(u^2)$ defect in the advection term, which is why these lattices suit low-Mach transport.

5.6 Magnetic induction — D2Q5, D3Q7

The magnetic field is carried by a distribution that is a *vector* at every link, with $b_\alpha = \sum_i g_i^\alpha$:

$$\boxed{g_i^{\mathrm{eq},\alpha} = w_i\left[b_\alpha + \frac{c_{i\beta}}{c_s^2}\,(u_\beta b_\alpha - b_\beta u_\alpha)\right]} \qquad \eta = c_s^2\Big(\frac{1}{\omega_\eta} - \frac{1}{2}\Big). \tag{25}$$

Its first two moments are $\sum_i g_i^{\mathrm{eq},\alpha} = b_\alpha$ and $\sum_i c_{i\beta} g_i^{\mathrm{eq},\alpha} = u_\beta b_\alpha - b_\beta u_\alpha$, which is exactly the antisymmetric flux of the induction equation. Only that first moment is required, which is why a five- or seven-velocity lattice suffices for the field while the flow runs on nine, nineteen or twenty-seven.

5.7 MHD fluid equilibrium — D2Q9, D3Q27

The Lorentz force enters through the equilibrium's second moment rather than as a body force. To (14) is added

$$\boxed{\delta f_i = \frac{w_i}{2c_s^4}\, M_{\alpha\beta}\big(c_{i\alpha}c_{i\beta} - c_s^2\delta_{\alpha\beta}\big), \qquad M_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2\delta_{\alpha\beta} - b_\alpha b_\beta} \tag{26}$$

so that $\sum_i c_{i\alpha}c_{i\beta} f_i^{\mathrm{eq}} = \rho u_\alpha u_\beta + (p + \frac{1}{2}|\boldsymbol{b}|^2)\delta_{\alpha\beta} - b_\alpha b_\beta$, while $\sum_i \delta f_i = 0$ and $\sum_i \boldsymbol{c}_i\,\delta f_i = \boldsymbol{0}$ leave mass and momentum untouched.

Dimensional form. In two dimensions $M_{\alpha\beta}\delta_{\alpha\beta} = |\boldsymbol{b}|^2 - |\boldsymbol{b}|^2 = 0$, so (26) reduces to the familiar

$$\delta f_i = \frac{w_i}{2c_s^4}\Big[\tfrac{1}{2}|\boldsymbol{c}_i|^2|\boldsymbol{b}|^2 - (\boldsymbol{c}_i\!\cdot\!\boldsymbol{b})^2\Big] \qquad \text{(2D only)}. \tag{27}$$

In three dimensions $M_{\alpha\beta}\delta_{\alpha\beta} = \frac{1}{2}|\boldsymbol{b}|^2 \neq 0$ and the trace subtraction is retained. This is not cosmetic: it is precisely the term whose projection onto the plane governs the quasi-two-dimensional reduction test of §10.9.3.

5.8 Summary: which equilibrium is used where

operator	lattices	equilibrium	where
BGK, TRT	D2Q9, D3Q27	second order, (14)	§5.2
MRT (raw moments)	D2Q9, D3Q27	moment space, (19), $\boldsymbol{u}_b = \mathbf{0}$	§5.4
central moments	D2Q9, D3Q27	moment space, (19), $\boldsymbol{u}_b = \boldsymbol{u}$	§5.4
MHD fluid (BGK)	D2Q9, D3Q27	(14) + (26)	§5.7
MHD central moments	D2Q9	(15) + (26); CMs (41)	§6.4
	D3Q27	(16) + (26); CMs (44)	§6.5
	(`-eq2`)	second order (14); CMs (40)	§6.4
scalar transport	D2Q5, D3Q7	first order, (21)	§5.5
magnetic induction	D2Q5, D3Q7	Dellar, (25)	§5.6
highest order each lattice admits: (15) D2Q9, (16) D3Q27			
regularised walls rebuild f_i from the same equilibrium, §8.1			

6 Collision operators

6.1 BGK

$$f_i^\star = f_i + \omega\,(f_i^{\mathrm{eq}} - f_i) + S_i, \qquad \nu = c_s^2\Big(\frac{1}{\omega} - \frac{1}{2}\Big). \tag{28}$$

6.2 TRT

Split each population into parts symmetric and antisymmetric about the opposite direction,

$$f_i^\pm = \tfrac{1}{2}\big(f_i \pm f_{\mathrm{opp}(i)}\big), \tag{29}$$

and relax them at different rates ω^+ (which sets the viscosity) and ω^- (free). The free rate buys the *magic parameter*

$$\Lambda = \Big(\frac{1}{\omega^+} - \frac{1}{2}\Big)\Big(\frac{1}{\omega^-} - \frac{1}{2}\Big). \tag{30}$$

At $\Lambda = 3/16$ the halfway bounce-back wall sits exactly midway between the fluid and solid nodes at *any* viscosity. BGK manages this only at the single value $\tau = \frac{1}{2} + \frac{\sqrt{3}}{4}$, because BGK is TRT with $\omega^- = \omega^+$.

The direction-ordering contract (7) pays off here: the pair loop is `for (i = 1; i < Q; i += 2)` with no lookup table. The equilibrium is evaluated in a separate flat loop first, because the pair loop reads f_i and f_{i+1} together and does not vectorise; leaving the equilibrium inside it cost more than the temporary array does (§13).

6.3 Moment-space collision: raw MRT and central moments

Both are one implementation. They differ only in the velocity at which the moment basis is constructed:

$$\boldsymbol{u}_b = \mathbf{0} \;\; \text{(raw moments, MRT)}, \qquad \boldsymbol{u}_b = \boldsymbol{u} \;\; \text{(central moments)}, \tag{31}$$

which is precisely the `central_moments` toggle of the symbolic generator.

6.3.1 Product basis (D2Q9, D3Q27)

A *product lattice* contains every velocity in $\{-1, 0, 1\}^D$ exactly once. For these the basis is a product of the same three one-dimensional functions on each axis,

$$\varphi_0(C) = 1, \qquad \varphi_1(C) = C, \qquad \varphi_2(C) = C^2 - c_s^2, \qquad C = c - u_b, \tag{32}$$

so the transform factorises into D successive one-dimensional passes rather than one $Q \times Q$ contraction:

$$\text{D3Q27: } 1458 \to 324 \text{ multiply–adds } (4.5\times), \qquad \text{D2Q9: } 162 \to 72. \tag{33}$$

Each pass, on values at $c = -1, 0, +1$, is

$$m_0 = g_- + g_0 + g_+, \qquad m_1 = (g_+ - g_-) - u\, m_0, \tag{34}$$

$$m_2 = (g_+ + g_-) - 2u(g_+ - g_-) + (u^2 - c_s^2)\, m_0, \tag{35}$$

with an exactly invertible counterpart. The factorised transform was checked against the dense contraction in exact rational arithmetic.

Why this basis. Basis (32) diagonalises the Maxwellian. The product-form equilibrium has *exactly one* nonzero moment in it:

$$k^{\mathrm{eq}}_{00\ldots} = \rho, \qquad k^{\mathrm{eq}}_{pqr} = 0 \ \text{ otherwise}, \tag{36}$$

verified symbolically for D2Q9 (0 of 9 deviating) and D3Q27 (0 of 27). No equilibrium populations are ever formed and no K^{eq} vector is stored, which is what collapses the collision to a handful of lines.

6.3.2 Relaxation and forcing

Writing $\delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}_b$, the equilibrium moments in the *stored* variable are

$$E_{pqr} = \rho\, Q_p(\delta u_x) Q_q(\delta u_y) Q_r(\delta u_z) - [\text{shifted}]\ A_p(u_{bx}) A_q(u_{by}) A_r(u_{bz}), \tag{37}$$

with $Q = \{1, \delta u, \delta u^2\}$ and $A = \{1, -u_b, u_b^2\}$ for the product basis (and the corresponding c_s^2-carrying triples for the monomial basis). The relaxation is then

$$\begin{aligned} \text{order } 0: &\quad \text{conserved} \\ \text{order } 1: &\quad k^\star = k + F_\alpha \\ \text{order } 2: &\quad \text{trace at } \omega_{\text{bulk}},\ \text{deviatoric and shear at } \omega \\ \text{order} \geq 3: &\quad k^\star = E. \end{aligned} \tag{38}$$

The Guo source has moments $\sum_i c_{i\alpha} F_i = F_\alpha$ and $\sum_i c_{i\alpha} c_{i\beta} F_i = u_\alpha F_\beta + u_\beta F_\alpha$. Transformed to the basis at $\boldsymbol{u}_b$ these become F_α and $(F_\beta\, \delta u_\alpha + F_\alpha\, \delta u_\beta)$. *For central moments $\delta\boldsymbol{u} = \mathbf{0}$, so the second-order force contribution vanishes identically* and the force enters only through the first moments — the well-known elegance of collision in the co-moving frame. For raw MRT it does not vanish and is applied with the usual $(1 - \omega/2)$ prefactor.

Bulk viscosity is an explicit `omega_bulk`, defaulting to ω.

6.4 MHD central moments

The hybrid scheme of De Rosis, Lévêque and Chahine [6] is implemented separately, because its equilibria are *not* the Maxwellian ones. Central moments are taken in that paper's non-orthogonal basis

$$\begin{aligned} &t_0 = 1, && t_1 = C_x, && t_2 = C_y, && t_3 = C_x^2 + C_y^2, \\ &t_4 = C_x^2 - C_y^2, && t_5 = C_x C_y, && t_6 = C_x^2 C_y, && t_7 = C_x C_y^2, \qquad t_8 = C_x^2 C_y^2, \end{aligned} \tag{39}$$

with $\boldsymbol{C} = \boldsymbol{c}_i - \boldsymbol{u}$, and relaxed toward

$$\begin{aligned} k_0^{\rm eq} &= \rho, & k_1^{\rm eq} = k_2^{\rm eq} = 0, \qquad k_3^{\rm eq} = \tfrac{2}{3}\rho, \\ k_4^{\rm eq} &= b_y^2 - b_x^2, & k_5^{\rm eq} = -b_x b_y, \\ k_6^{\rm eq} &= -\rho u_x^2 u_y + \tfrac{u_y}{2}(b_x^2 - b_y^2) + 2u_x b_x b_y, \\ k_7^{\rm eq} &= -\rho u_x u_y^2 + \tfrac{u_x}{2}(b_y^2 - b_x^2) + 2u_y b_x b_y, \\ k_8^{\rm eq} &= \tfrac{\rho}{9}(27u_x^2 u_y^2 + 1) + \tfrac{u_x^2 - u_y^2}{2}(b_x^2 - b_y^2) - 4u_x u_y b_x b_y, \end{aligned} \tag{40}$$

with $\omega_4 = \omega_5 = \omega_\nu$, ω_3 setting the bulk viscosity and $\omega_6 = \omega_7 = \omega_8 = 1$. k_0, k_1, k_2 are collision invariants.

Raising the equilibrium order in two dimensions. Substituting the product form (15) for (14) leaves k_0–k_5 of (40) untouched and removes precisely the ρ-dependent terms from the rest:

$$k_6^{\rm eq} = \tfrac{u_y}{2}(b_x^2 - b_y^2) + 2u_x b_x b_y, \qquad k_7^{\rm eq} = \tfrac{u_x}{2}(b_y^2 - b_x^2) + 2u_y b_x b_y, \qquad k_8^{\rm eq} = \tfrac{\rho}{9} + \tfrac{u_x^2 - u_y^2}{2}(b_x^2 - b_y^2) - 4u_x u_y b_x b_y. \tag{41}$$

This is the default; (40) is selected by `-eq2` and is what the validation of §10.9 against the published Table 1 uses.

These equilibria are deliberately non-Maxwellian. They are the central moments of the *second-order truncated* equilibrium plus the Maxwell term, not of a product-form equilibrium. The terms $-\rho u_x^2 u_y$ in k_6 and $3\rho u_x^2 u_y^2$ in k_8 are exactly the Galilean defects of the truncation. Reproducing the published scheme means reproducing them rather than improving them away, which is why this is a separate operator rather than a configuration of §6.3.

Verification. Equation (40) was not taken on trust. The test suite constructs the paper's equilibrium (67) directly, contracts its central moments by summation, and compares: all nine agree to 1.1×10^{-16}, which also confirms the sign convention on $\boldsymbol{b}$.

6.5 Three dimensions: D3Q27, and the effect of equilibrium order

Reference [6] prescribes D3Q27 for the velocity field (with D3Q7 for the magnetic field) in three dimensions. The operator is extended accordingly. Writing $\boldsymbol{C} = \boldsymbol{c}_i - \boldsymbol{u}$ and

$$k_{pqr} = \sum_i f_i\, C_x^p\, C_y^q\, C_z^r, \qquad M_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2 \delta_{\alpha\beta} - b_\alpha b_\beta, \tag{42}$$

the equilibrium central moments follow from contracting the three-dimensional equilibrium — hydrodynamic part plus the Maxwell term (26) — against the monomial basis. They were obtained symbolically in exact rational arithmetic, and depend on which hydrodynamic equilibrium is used.

With the second-order truncation (14). Orders two and three close in a form that holds for every index combination,

$$k^{\mathrm{eq}}_{\alpha\beta} = \rho c_s^2 \delta_{\alpha\beta} + M_{\alpha\beta}, \qquad k^{\mathrm{eq}}_{\alpha\beta\gamma} = -\rho\, u_\alpha u_\beta u_\gamma - \big(u_\alpha M_{\beta\gamma} + u_\beta M_{\alpha\gamma} + u_\gamma M_{\alpha\beta}\big), \tag{43}$$

and 24 of the 27 moments are nonzero. Setting $u_z = b_z = 0$ reduces (43) and its fourth-order companions to (40) exactly, moment for moment — the three-dimensional scheme is a genuine extension of the published one, not a different model that happens to agree in the bulk.

With the sixth-order equilibrium (16). Because (18) holds, the hydrodynamic contribution collapses to the Maxwellian and *every ρ-dependent term above second order disappears*:

$$\boxed{\begin{aligned}
k^{\mathrm{eq}}_{000} &= \rho, \qquad k^{\mathrm{eq}}_{\alpha} = 0,\\
k^{\mathrm{eq}}_{\alpha\beta} &= \rho c_s^2 \delta_{\alpha\beta} + M_{\alpha\beta},\\
k^{\mathrm{eq}}_{\alpha\beta\gamma} &= -\big(u_\alpha M_{\beta\gamma} + u_\beta M_{\alpha\gamma} + u_\gamma M_{\alpha\beta}\big),\\
k^{\mathrm{eq}}_{aacc} &= \tfrac{\rho}{9} + \tfrac{1}{3} b_d^2 + u_c^2 M_{aa} + u_a^2 M_{cc} + 4 u_a u_c M_{ac},\\
k^{\mathrm{eq}}_{aacd} &= u_c u_d M_{aa} + \big(\tfrac{1}{3} + u_a^2\big) M_{cd} + 2 u_a u_d M_{ac} + 2 u_a u_c M_{ad},\\
k^{\mathrm{eq}}_{acc\,dd} &= -\tfrac{1}{3} u_a b_a^2 - \tfrac{2}{3}\big(u_c M_{ac} + u_d M_{ad}\big) - 2 u_c u_d^2 M_{ac} - 2 u_c^2 u_d M_{ad}\\
&\quad - 4 u_a u_c u_d M_{cd} - u_a u_d^2 M_{cc} - u_a u_c^2 M_{dd},\\
k^{\mathrm{eq}}_{222} &= \tfrac{\rho}{27} + \tfrac{1}{18}|\boldsymbol{b}|^2 + \tfrac{1}{3}\sum_{\alpha} u_\alpha^2 b_\alpha^2 + \tfrac{4}{3}\sum_{\alpha<\beta} u_\alpha u_\beta M_{\alpha\beta}\\
&\quad + 4\big(u_x u_y u_z^2 M_{xy} + u_x u_y^2 u_z M_{xz} + u_x^2 u_y u_z M_{yz}\big)\\
&\quad + u_y^2 u_z^2 M_{xx} + u_x^2 u_z^2 M_{yy} + u_x^2 u_y^2 M_{zz},
\end{aligned}} \tag{44}$$

where (a, c, d) denotes a permutation of (x, y, z). Comparing with the second-order case, the terms lost are exactly $-\rho u_\alpha u_\beta u_\gamma$ at order three, $3\rho u_a^2 u_c^2$ and $3\rho u_a^2 u_c u_d$ at order four, and so on: the Galilean defects, and nothing else. What remains is the Maxwell stress, which is not a defect but the physics.

Implementation and verification. Rather than evaluate 27 closed forms — orders five and six do not factorise compactly — the operator builds the equilibrium populations and transforms them with the same monomial central-moment transform used for the populations themselves. The order of the hydrodynamic part is a compile-time switch, defaulting to (16); the second-order form is retained so the published results of [6] stay reproducible. The test suite checks, at *both* orders: the monomial transform against a direct contraction (2.2×10^{-16}); the closed forms (43) and (44) (5.6×10^{-17}); the reduction to the two-dimensional scheme at $u_z = b_z = 0$; and conservation of mass and all three momentum components.

Relaxation. As in two dimensions, the trace $k_{200} + k_{020} + k_{002}$ relaxes at ω_{bulk}, its five deviatoric partners at ω_ν, and everything of order three and above goes straight to equilibrium. Note that in three dimensions the trace carries a magnetic contribution that is absent in two,

$$k^{\mathrm{eq}}_{200} + k^{\mathrm{eq}}_{020} + k^{\mathrm{eq}}_{002} = \rho + \tfrac{1}{2}|\boldsymbol{b}|^2, \tag{45}$$

because $M_{\alpha\alpha} = \frac{1}{2}|\boldsymbol{b}|^2$ in 3D but vanishes in 2D — the same trace term discussed in §5.7.

7 Forcing

7.1 Guo

$$\boldsymbol{u} = \frac{1}{\rho}\Big(\sum_i \boldsymbol{c}_i f_i + \tfrac{1}{2}\boldsymbol{F}\Big), \qquad S_i = \Big(1 - \frac{\omega}{2}\Big) w_i \left[\frac{\boldsymbol{c}_i - \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_i \cdot \boldsymbol{u})\, \boldsymbol{c}_i}{c_s^4}\right] \cdot \boldsymbol{F}. \tag{46}$$

The $(1-\omega/2)$ prefactor is applied *per relaxed mode*, not once: TRT and the moment operators relax different modes at different rates, so only BGK can fold a single prefactor in. The raw source is exposed separately for that reason.

7.2 High-order Hermite forcing, and why it is already here

The Guo term (46) is the body force expanded in Hermite polynomials and truncated at second order. Nothing stops the expansion being carried further, and on these lattices it can be taken as far as the equilibrium goes: fourth order on D2Q9, sixth on D3Q27. Writing $\mathcal{H}^{(n)}$ for the Hermite tensors in $\boldsymbol{c}_i$ and building the coefficients by the product rule — replacing one $\boldsymbol{u}$ slot by $\boldsymbol{F}$ at a time,

$$a^{(1)}_{\alpha} = F_\alpha, \quad a^{(2)}_{\alpha\beta} = F_\alpha u_\beta + u_\alpha F_\beta, \quad a^{(3)}_{\alpha\beta\gamma} = F_\alpha u_\beta u_\gamma + u_\alpha F_\beta u_\gamma + u_\alpha u_\beta F_\gamma, \ \dots \tag{47}$$

— the source term is

$$F_i = w_i \sum_{n\geq 1} \frac{1}{n!\, c_s^{2n}}\, a^{(n)}_{\alpha_1\dots\alpha_n}\, \mathcal{H}^{(n)}_{\alpha_1\dots\alpha_n}(\boldsymbol{c}_i). \tag{48}$$

Which components survive. Not every Hermite component may be kept. Two rules, both forced by the lattice rather than chosen:

- Components with an odd exponent ≥ 3 on one axis *vanish identically.* On D2Q9, $c_x^3 = c_x$ and $1/c_s^2 = 3$ give $\mathcal{H}^{(3)}_{xxx} = \hat{c}_x^3 - 3\hat{c}_x \equiv 0$. They may be kept or dropped freely.
- Components with an *even* exponent ≥ 4 on one axis are nonzero but *not independent*: $\mathcal{H}^{(4)}_{xxxx} = -3\,\mathcal{H}^{(2)}_{xx}$. Including them injects a spurious $O(Fu^3)$ term into the second moment and destroys $k_{20} = k_{02} = 0$. They must be omitted.

What remains is exactly the set of exponent triples in $\{0, 1, 2\}$ — the same representable set the equilibrium uses.

The result, and it is simple. Transforming (48) to *monomial* central moments $k_{pqr} = \sum_i F_i\, C_x^p C_y^q C_z^r$ collapses to one rule. If exactly one axis a carries exponent 1 and m other axes carry exponent 2, then

$$\boxed{k^{\text{force}} = c_s^{2m} F_a} \qquad \text{and every other moment is exactly zero.} \tag{49}$$

On D2Q9 that is $[0, F_x, F_y, 0, 0, 0, c_s^2 F_y, c_s^2 F_x, 0]$ in the ordering $(00, 10, 01, 20, 02, 11, 21, 12, 22)$. On D3Q27 it is three entries F_a, six at $c_s^2 F_a$, three at $c_s^4 F_a$, and fifteen exact zeros. In the co-moving frame the force touches only the moments that are odd in exactly one axis — which is the whole reason for working there.

And the code already does this. It is tempting to read (49) as saying that an operator which adds $\boldsymbol{F}$ only to the first-order moments, as §6.3 does, is missing the c_s^{2m} entries. It is not. `ProductBasis` stores $\varphi_2(C) = C^2 - c_s^2$ per axis, so expanding $C^2 = \varphi_2 + c_s^2$,

$$k^{\text{monomial}}_{122} = \tilde{k}_{122} + c_s^2 \tilde{k}_{120} + c_s^2 \tilde{k}_{102} + c_s^4 \tilde{k}_{100}, \tag{50}$$

and with only $\tilde{k}_{100} = F_x$ populated the monomial value is $c_s^4 F_x$, exactly as required — one factor of c_s^2 per squared axis, delivered by the basis function itself. The momentum source alone reproduces all of (49).

This was established the hard way. The third-order terms were added explicitly on top, and the operator then injected $1.5\, c_s^2 F_y$ where $c_s^2 F_y$ is correct — a clean 50% overshoot from double counting. No change to the collision operator was needed or kept. `validation/forcing_cm.cpp` checks both representations of the same populations on both lattices: monomial must show the full c_s^{2m} pattern, Hermite must show the force in the first-order slots and nowhere else. Worst deviation 5.6×10^{-13}.

7.3 Boussinesq buoyancy

$$\boldsymbol{F}(n) = \rho_0\, \boldsymbol{g}\, \beta \left(T(n) - T_0\right), \tag{51}$$

delivered through the same Guo machinery. Because this force varies in space, the collision interface carries a node index; that hook is also what the Lorentz force uses.

8 Boundary conditions

Boundary conditions are alternative collision operators applied to flagged cells. This gives one branch per *node* rather than one per direction, keeps the population loads unconditional and coalesced, and generalises to any wall model.

condition	rule	wall location
no-slip (halfway bounce-back)	$f_i^{\text{out}} = f_{\text{opp}(i)}^{\text{in}}$	midway
adiabatic scalar	$g_i^{\text{out}} = g_{\text{opp}(i)}^{\text{in}}$	midway
Dirichlet scalar (anti-bounce-back)	$g_i^{\text{out}} = -g_{\text{opp}(i)}^{\text{in}} + 2w_i\,(T_w - T_{\text{ref}})$	midway

There is an asymmetry worth stating. Bounce-back is the *identity* on Esoteric Pull's storage (§3.2), so those cells are skipped entirely. Anti-bounce-back flips a sign and adds a source, so those cells must be processed at every step. Both still write back into the same two slots they read, so neither disturbs the in-place scheme.

All three conditions above put the wall *midway* between nodes. The three that follow put it *on* the node. That difference is not a detail of accuracy but of geometry: a channel of N nodes is N lattice units wide under bounce-back and $N-1$ under an on-node condition, and mixing the two families in one problem means the fluid and the field disagree about where the wall is.

8.1 Regularised velocity walls

Boundary condition BC3 of Latt, Chopard, Malaspinas, Deville and Michler [1]. *Every* population on the node is replaced, using only ρ, the imposed $\boldsymbol{u}$, and the non-equilibrium stress:

$$\boxed{g_i = f_i^{\text{eq}}(\rho, \boldsymbol{u}) + \frac{w_i}{2c_s^4}\, \boldsymbol{Q}_i : \Pi^{(1)}, \qquad \boldsymbol{Q}_i = \boldsymbol{c}_i\boldsymbol{c}_i - c_s^2 \boldsymbol{I}.} \tag{52}$$

Note what (52) does *not* contain: ω. The reconstruction is a projection of a given tensor onto the lattice, so it serves BGK, TRT, MRT and the central-moment operators without change.

Density. The paper's closure is written for one particular figure and velocity numbering. It is implemented here geometrically, which is numbering independent and therefore survives the Esoteric Pull ordering: with $\hat{\boldsymbol{n}}$ the *outward* normal, the known populations are those with $\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} > 0$, and

$$\rho = \frac{1}{1 + \boldsymbol{u} \cdot \hat{\boldsymbol{n}}} \Big(2 \sum_{\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} > 0} f_i + \sum_{\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} = 0} f_i \Big). \tag{53}$$

Stress. The unknown populations are given, *temporarily*, a bounce-back of their off-equilibrium part, and $\Pi^{(1)} = \sum_i \boldsymbol{c}_i\boldsymbol{c}_i(f_i - f_i^{\mathrm{eq}})$ is evaluated from that. The paper is explicit that this may not be used as the boundary condition itself — it cannot enforce $\boldsymbol{u}$ exactly — and it exists here only to give $\Pi^{(1)}$ a value.

Which populations are unknown. Not a half-space. A dot-product test $\boldsymbol{c}_i\cdot\hat{\boldsymbol{n}} < 0$ is correct on a straight wall and *wrong* on a corner: at the corner $(0,0)$ of a box, both $(1,-1)$ and $(-1,1)$ are unknown, each having its source node outside a different wall. The code therefore carries a per-node bitmask, built once from the geometry: bit i is set when the source $\boldsymbol{p} - \boldsymbol{c}_i$ lies outside the fluid. Where *both* members of a pair are unknown — possible only on corners and edges — those populations are set to equilibrium, contributing nothing to $\Pi^{(1)}$, which is the only choice that invents no stress.

Corners and edges. Equation (53) needs a single normal, which a corner does not have, so ρ is extrapolated to second order along one wall from two straight-wall neighbours — never reaching into the bulk, so no density field is required. For $\Pi^{(1)}$ there are two routes, and the default is the second:

- *local* — the same closure as a straight wall. Operator-agnostic, but a 2D corner offers only three streamed directions to build a stress from.
- *finite-difference* — the route of [1], Sec. V: $\Pi^{(1)}$ from second-order one-sided velocity gradients through

$$\Pi^{(1)} = -\frac{2c_s^2}{\omega}\,\rho\,\boldsymbol{S}, \qquad \boldsymbol{S} = \tfrac{1}{2}\big[\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T\big]. \tag{54}$$

Neighbour velocities are taken from the imposed value when the neighbour is itself a wall node — its own unknowns are not yet fixed, so reading its populations would be wrong — and from its populations otherwise.

This is where the condition acquires a collision dependence. Equation (54) is a constitutive law, and its coefficient is the viscosity $\nu = c_s^2(1/\omega - 1/2)$. Inferring stress from strain rate therefore *requires* naming the shear relaxation rate, whereas measuring it from the populations does not: the operator's effect is already carried by the data. The rate is read from the operator (`omega`, or `omega_p` for TRT, whose even rate sets the viscosity), never supplied by the caller; an operator exposing none cannot select this route at all, which removes by construction the possibility of silently evaluating (54) with a placeholder $\omega = 1$ and imposing the wrong viscosity at the wall.

Body forces. Guo forcing defines $\rho\boldsymbol{u} = \sum_i f_i\boldsymbol{c}_i + \boldsymbol{F}/2$, so a reconstruction must satisfy $\sum_i \boldsymbol{c}_i g_i = \rho\boldsymbol{u}_w - \boldsymbol{F}/2$. Equation (52) delivers $\rho\boldsymbol{u}_w$, because $\sum_i w_i\boldsymbol{c}_i\boldsymbol{Q}_i = \boldsymbol{0}$. Two corrections follow from the forced Chapman–Enskog expansion and both are applied by default.

First, the expansion gives (dropping $O(\mathrm{Ma}^3)$)

$$\sum_i \boldsymbol{c}_i f_i^{(1)} = -\frac{\boldsymbol{F}}{2}, \qquad \Pi^{(1)} = -\frac{2\rho c_s^2}{\omega}\boldsymbol{S} - \frac{1}{2}(\boldsymbol{u}\boldsymbol{F} + \boldsymbol{F}\boldsymbol{u}), \tag{55}$$

the second term vanishing at a no-slip wall. The second-order Hermite reconstruction of $f^{(1)}$ consistent with (55) therefore needs its own first-moment term, which $\boldsymbol{Q}_i : \Pi$ cannot supply:

$$g_i \;\to\; g_i \;-\; \frac{w_i}{2c_s^2}\,\boldsymbol{c}_i\cdot\boldsymbol{F}. \tag{56}$$

Second — and this is the larger effect — the *measurement* of $\Pi^{(1)}$ is biased. Since $f^{(1)}$ carries the odd part $-\frac{w_i}{2c_s^2}\boldsymbol{c}_i\cdot\boldsymbol{F}$, bounce-back reverses its sign, so each unknown is wrong by $(w_i/c_s^2)\,\boldsymbol{c}_i\cdot\boldsymbol{F}$. Contracting $\boldsymbol{c}_i\boldsymbol{c}_i$ against that error would cancel over the full velocity set, but the unknowns form a *half-space* and it does not. On D2Q9 at a flat wall:

force direction	$\delta\Pi_{xx}$	$\delta\Pi_{yy}$	$\delta\Pi_{xy}$
tangential	0	0	$F/6$
normal	$F/6$	$F/2$	0

The tensors are different, which is why no correction of the form $\boldsymbol{u}_w \mp \boldsymbol{F}/2\rho$ can serve both cases: a tangential force produces spurious *shear*, a normal force spurious *normal* stress. The scaffold is therefore given the correct odd part rather than the reversed one.

What remains is not a force effect. After (56) and the scaffold repair, a residual wall offset survives for $\omega \neq 1$, obeying

$$\text{offset} \times W = \frac{4}{3}\,\frac{\omega-1}{\omega^2} \qquad \Longleftrightarrow \qquad \delta u_{\text{slip}} = -\frac{2}{3}\,\frac{\omega-1}{\omega^2}\,\frac{\partial^2 u_\parallel}{\partial n^2}. \tag{57}$$

This is a *curvature*-induced slip, not a force term. It is independent of $\boldsymbol{F}$ to five digits across an eightfold range of forcing; it vanishes identically for a linear profile, so Couette flow is exact to 4×10^{-15} at every τ from 0.6 to 2.0; and it vanishes at $\omega = 1$, consistent with the factor $(1-\omega)$ by which collision erases any error in the emitted stress. It is a pre-existing consequence of truncating $f^{(1)}$ at second Hermite order, which is what "regularised" means, and no correction for it is applied. The law (57) is empirical: it was fitted at two values of τ and then *predicted* three held-out values to within 0.3%, but the coefficient 4/3 has not been derived.

Departure from the paper. Reference [1] uses BC5, Eq. (48), on corners and edges; the form implemented here is BC4, Eq. (46). The two differ only by terms nonlinear in $\boldsymbol{u}$, which — as [1] states — "cancel for symmetry reasons during the evaluation of the tensor $\Pi^{(1)}$", and the asymptotic dynamics depends on $\Pi^{(1)}$ alone. The two therefore impose the same stress and differ only in ghost-population detail.

8.2 Outflow: constant back-pressure

An open outlet needs a different closure from a wall: the velocity there is not known, it is an output of the interior flow. The condition implemented reuses the regularised reconstruction of §8.1 and differs only in where $(\rho, \boldsymbol{u})$ come from.

Why the density must be imposed. The obvious choice — copy both ρ and $\boldsymbol{u}$ from the upstream neighbour, a zero-gradient outlet — is *ill posed* in this scheme, and not marginally so. The regularised nodes overwrite every population from $(\rho, \boldsymbol{u}, \Pi^{(1)})$, so they are not mass conserving; with a velocity inlet injecting mass and no condition fixing the pressure anywhere, the total mass has no equilibrium. Run on the channel of §11.6 it settles at $\rho \approx 181$ with a density ramp along x and a centreline velocity a factor two below the analytic profile. The outlet must therefore fix the pressure:

$$\rho(\text{outlet}) = \rho_0. \tag{58}$$

Normal velocity. With ρ known, the wall closure (53) inverts. Partitioning the populations by the sign of $\boldsymbol{c_i} \cdot \hat{\boldsymbol{n}}$ with $\hat{\boldsymbol{n}}$ the *outward* normal, and writing $\Sigma_+ = \sum_{\boldsymbol{c_i}\cdot\hat{\boldsymbol{n}}>0} f_i$ (streamed from the interior, hence known) and $\Sigma_0 = \sum_{\boldsymbol{c_i}\cdot\hat{\boldsymbol{n}}=0} f_i$,

$$\rho\,(1 + \boldsymbol{u}\cdot\hat{\boldsymbol{n}}) = 2\Sigma_+ + \Sigma_0 \qquad \Longrightarrow \qquad \boxed{\boldsymbol{u}\cdot\hat{\boldsymbol{n}} = \frac{2\Sigma_+ + \Sigma_0}{\rho} - 1} \tag{59}$$

which is the same relation as the wall's, read in the other direction. The outlet therefore passes whatever flux the interior sends it rather than prescribing one, and the condition is self-regulating.

Tangential velocity. The node's own populations carry no inflow directions, so the tangential components are taken from the upstream neighbour (zero gradient, first order). A second-order variant — linear extrapolation from two upstream nodes, `set_outflow_order(2)` — is implemented and is *not* used, because it was measured to be no better: worse on coarse grids (1.388×10^{-3} against 1.160×10^{-3} at $L_y = 41$) and indistinguishable on the finest (6.184×10^{-6} against 6.192×10^{-6} at $L_y = 641$). §11.6 shows why — the outlet is not what limits the order there.

Only the $+x$ face is implemented (`NrmOutXp`). Nothing in (59) is specific to that direction; the other five are a matter of extending `upstream_of` and `normal_of`.

The magnetic counterpart, and its defect. The same idea is available on the induction lattices (`MagOutXp`): the node takes $\boldsymbol{b}$ from its upstream neighbour instead of an imposed value, and is then closed by the moment condition (63) exactly as a Dirichlet magnetic wall is. It reuses the field pass rather than the collision pass – the upstream neighbour's populations are read where $\boldsymbol{b}$ is computed, which is race free because that kernel only reads populations, and the node's own $\boldsymbol{b}$ then serves as the target when the moment is imposed.

This condition is not validated and has a measured defect. On the inlet-driven Hartmann flow of §10.10 — whose exact solution is independent of the streamwise coordinate, so that any x variation in the answer is boundary error and nothing else – it drives $\boldsymbol{b}$ about 6% high over the last ten or so nodes, and the overshoot decays upstream only slowly:

L_x	b/b_{exact} along the channel at $L_y = 65$
21	runs $1.000 \to 1.054$ across the *whole* domain
121	flat plateau at 1.005 in the bulk, rising to 1.061 at the end
241	plateau reached well before the sampling station

The effect on a score is large: at $L_y = 129$ the error in $\boldsymbol{b}$ falls from 3.367×10^{-3} to 9.793×10^{-4} purely by moving the outlet from $L_x = 121$ to $L_x = 241$. It is specifically the magnetic outlet and not the fluid one — pinning $\boldsymbol{b}$ analytically at the outlet instead gives errors agreeing to three significant figures with the zero-gradient case at $L_y = 17, 33, 65$, so the two differ only through that contamination. The cause is not identified. The two natural suspects are the interaction with $\nabla\cdot\boldsymbol{b} = 0$, which this code does not preserve to machine precision anyway, and the fact that a pure Neumann condition does not constrain the field level at all — the same class of ill-posedness that made a zero-gradient *fluid* outlet drive ρ to 181. Until it is understood, keep the outlet far from anything being measured.

8.3 Forcing: every component, every axis, and a fully 3D case

§10.1 drives its channel along x with $\boldsymbol{F} = (G, 0, 0)$. That exercises $k_{12} = c_s^2 F_x$ from (49) and never touches $k_{21} = c_s^2 F_y$, so on its own it validates half the forcing. Three tests close that.

Rotating the channel, D2Q9. Walls x-normal and the force along $+y$ swaps which component is live. Both orientations are exact at the central-moment magic point, and — since D2Q9 is invariant under a quarter turn — they must agree with each other, which no single orientation can check.

	$\tau = 0.6$	$\tau = 7/8$	$\tau = 1.2$
along x (tests k_{12})	3.9225×10^{-3}	3.06×10^{-14}	4.6356×10^{-3}
along y (tests k_{21})	3.9225×10^{-3}	3.14×10^{-14}	4.6356×10^{-3}
|difference|	5.3×10^{-14}	6.1×10^{-15}	1.1×10^{-14}

$H = 16$. Off-magic the two agree to 10^{-13}, which is rotational isotropy of the forcing, equilibrium and streaming together.

All three axes, D3Q27. Walls normal to one axis, force along the next, periodic in the third; the three cyclic combinations put every axis in both roles. At $H = 16$ they give 2.99×10^{-14}, 3.68×10^{-14}, 3.12×10^{-14} at $\tau = 7/8$, and off-magic all three are *bit-identical* — 3.9225×10^{-3} at $\tau = 0.6$ and 4.6356×10^{-3} at $\tau = 1.2$, the same values D2Q9 gives, since a plane channel is quasi-one-dimensional and D3Q27 reduces to D2Q9 exactly on it.

A fully three-dimensional forced flow. A plane channel varies along one axis only, so it never excites a moment with two squared axes and barely touches the c_s^4 group. This case does. Periodic cube, no walls, $F_x = F_0 \sin(ky)\sin(kz)$ with $k = 2\pi/N$, whose steady solution is *exact* for the full Navier–Stokes equations,

$$u_x = \frac{F_0 \sin(ky)\sin(kz)}{2\rho\nu k^2}, \qquad u_y = u_z = 0, \tag{60}$$

because u_x does not depend on x, so $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$ vanishes identically and the pressure stays uniform. The nonlinear term is not small here; it is zero.

N	$\tau = 0.8$		$\tau = 1.2$	
	rel L_2	order	rel L_2	order
8	7.4948×10^{-2}	—	8.1933×10^{-2}	—
16	2.0117×10^{-2}	1.897	2.0529×10^{-2}	1.997
32	5.1128×10^{-3}	1.976	5.1381×10^{-3}	1.998
64	1.2834×10^{-3}	1.994	1.2850×10^{-3}	2.000

Second order, and the L_2 error *equals* the peak error at every entry: the profile shape is right to round-off and all the error is amplitude.

Two magic points, and they are not the same one. That amplitude error changes sign with τ, so some viscosity makes it vanish. Scanning τ from 0.6 to 1.4 at $N = 32$ locates the crossing at $\tau = 0.9994$, and the whole family collapses onto

$$\frac{\Delta A}{A} = 2c_s^2 k^2(\tau - 1) + O(k^4), \tag{61}$$

with measured-to-model ratios running $0.9946 \to 0.9999$ as the grid refines, across $\tau \in [0.6, 1.4]$ and $N \in [8, 64]$. So the forced scheme has two distinct magic points and no single τ gives both: $\tau = 7/8$ puts the *wall* exactly on the node, a boundary property; $\tau = 1$ makes the *bulk* amplitude exact, a dispersion property. Which one limits a given run depends on whether it is wall-dominated or bulk-dominated.

The tests can fail. An exactness claim is only worth as much as its sensitivity, so the third-order force moments were deliberately perturbed by +50% and the tests re-run. The D2Q9 channel degrades from 8.44×10^{-14} to 6.69×10^{-4} and its wall slip from 7.4×10^{-14} to -3.9×10^{-3}; the transverse case, where $F_x = 0$ so only k_{21} is live, degrades from 3.14×10^{-14} to 2.67×10^{-3}. Ten orders of magnitude in both. These are not tests that pass regardless.

8.4 Moment-based walls for the scalar and magnetic fields

Dellar's moment conditions [2], Eqs. (13a)–(13b). The fields carried by the advection–diffusion lattices are *zeroth moments*,

$$B_\alpha = \sum_i g_{\alpha i}, \qquad T = \sum_i h_i, \tag{62}$$

so imposing a boundary value is one linear equation in the unknown populations. On a cross lattice a straight wall leaves *exactly one* unknown — the direction pointing into the domain along the normal — and that single equation closes the system uniquely and exactly:

$$\boxed{g_{x1} = B_0 - \big(g_{x0} + g_{x2} + g_{x3} + g_{x4}\big), \qquad g_{y1} = -\big(g_{y0} + g_{y2} + g_{y3} + g_{y4}\big).} \tag{63}$$

There is no closure assumption and no free parameter. The scalar case is identical with $\sum_i h_i = T_w - T_{\mathrm{ref}}$, the shift being the storage reference of §5.5.

Why B may simply be imposed. Maxwell's equations make both the normal and the tangential components of $\boldsymbol{b}$ continuous across the wall, so $\boldsymbol{b}$ takes its external applied value there. The velocity enjoys no such freedom, which is why $\boldsymbol{u}$ needs the machinery of §8.1 and $\boldsymbol{b}$ does not.

Corners. More than one unknown, and the single moment no longer closes the system. The deficit is then shared in proportion to the lattice weights: the moment is still reproduced exactly, only its distribution among the unknowns is a choice, and the weight-proportional one introduces no directional preference. This is a fallback, not part of the published method.

A trap worth recording. The field kernels compute B_α and T as $\sum_i$ of the streamed populations. At a moment wall those still contain the *unfixed* inward direction, so their sum is not the field. Reading it anyway feeds a wrong $\boldsymbol{b}$ into the magnetic equilibrium and the computation diverges — observed here at 10^{150} within a few hundred steps. The field at such a node is the imposed value by construction and is reported as such. The same applies to ρ and $\boldsymbol{u}$ at a regularised wall.

condition	fields	rule	wall location
regularised	$\rho, \boldsymbol{u}$	(52) with (53)	on the node
moment (magnetic)	$\boldsymbol{b}$	(63)	on the node
moment (scalar)	T	$\sum_i h_i = T_w - T_{\mathrm{ref}}$	on the node

9 Coupled modules

9.1 Thermal transport

A passive scalar is carried by a second distribution set on its own lattice, with

$$T = \sum_i g_i, \qquad g_i^{\mathrm{eq}} = w_i T\Big(1 + \frac{\boldsymbol{c}_i \cdot \boldsymbol{u}}{c_s^2}\Big), \qquad D = c_s^2\Big(\frac{1}{\omega} - \frac{1}{2}\Big). \tag{64}$$

The velocity is an *input*, not something recovered from these populations, which is why the scalar has its own collision concept.

The first-order truncation in (64) is deliberate. D2Q5 and D3Q7 have an isotropic second moment but not an isotropic fourth-order one, and on D3Q7 every velocity has a single nonzero component, so $(\boldsymbol{c}_i\boldsymbol{u})^2$ reduces to $c_\alpha^2 u_\alpha^2$ and the cross terms of the $\boldsymbol{uu}$ tensor cannot be represented at all. A second-order term would add an anisotropic error rather than accuracy. The cost is an $O(u^2)$ defect in the advection term, which is why these lattices suit low-Mach transport — exactly the regime Boussinesq convection occupies.

Architectural significance. This module was the first test of the composition the design was built around. D3Q7 has seven populations to the fluid's nineteen, a different sound speed, a different collision concept and different boundary conditions — and nothing in `FluidSolver`, the lattices, the streaming schemes or the moment operators required modification. The only interface addition was the node index on the collision.

9.2 Magnetohydrodynamics

Induction. Following Dellar [5], the magnetic field is carried by a *vector-valued* distribution g_i^α, with

$$b_\alpha = \sum_i g_i^\alpha, \qquad g_i^{\mathrm{eq},\alpha} = W_i\left[b_\alpha + \frac{c_{i\beta}}{\theta^2}(u_\beta b_\alpha - b_\beta u_\alpha)\right], \qquad \eta = \theta^2\Big(\frac{1}{\omega_\eta} - \frac{1}{2}\Big). \tag{65}$$

Taking moments gives $\sum_i g_i^{\mathrm{eq}} = b_\alpha$ and $\sum_i c_{i\beta} g_i^{\mathrm{eq},\alpha} = u_\beta b_\alpha - b_\beta u_\alpha$, which is exactly the antisymmetric flux of (4). Only that first moment is needed, so the magnetic lattice can be much smaller than the fluid's: D2Q5 or D3Q7 suffices. Running the field on a different lattice from the flow is the point, not an economy.

Lorentz force. The momentum equation gains $-\nabla\cdot(\boldsymbol{bb} - \frac{1}{2}|\boldsymbol{b}|^2 I)$. This is *not* applied as a body force, which would require derivatives of $\boldsymbol{b}$ and lose an order. The fluid equilibrium is instead given the correct second moment directly,

$$\sum_i c_{i\alpha} c_{i\beta} f_i^{\mathrm{eq}} = \rho u_\alpha u_\beta + \Big(p + \tfrac{1}{2}|\boldsymbol{b}|^2\Big)\delta_{\alpha\beta} - b_\alpha b_\beta, \tag{66}$$

achieved by adding to the hydrodynamic equilibrium

$$\delta f_i = \frac{w_i}{2c_s^4} M_{\alpha\beta}(c_{i\alpha}c_{i\beta} - c_s^2\delta_{\alpha\beta}), \qquad M_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2\delta_{\alpha\beta} - b_\alpha b_\beta. \tag{67}$$

Since $\sum_i \delta f_i = 0$ and $\sum_i \boldsymbol{c}_i\,\delta f_i = \mathbf{0}$, this term perturbs only the stress and leaves the shifted-storage bookkeeping untouched. In two dimensions $M_{\alpha\beta}\delta_{\alpha\beta} = 0$, so (67) reduces to the familiar $\frac{w_i}{2c_s^4}\big[\frac{1}{2}|\boldsymbol{c}_i|^2|\boldsymbol{b}|^2 - (\boldsymbol{c}_i\cdot\boldsymbol{b})^2\big]$; in three dimensions the trace subtraction matters and is retained.

9.3 Coupling order

Two-way coupling must be evaluated *simultaneously.* If the fluid is stepped before the magnetic field is refreshed, it collides against $\boldsymbol{b}(t-1)$. That is a first-order splitting error, and — crucially — it does *not* vanish under refinement: with diffusive scaling the ratio

$$\frac{\omega^2\,\Delta t}{\nu k^2} \tag{68}$$

is independent of N, so it appears as a damping offset that survives every grid refinement.

This was found in practice. The shear Alfvén damping error *grew* with resolution, $1.55 \times 10^{-2} \to 2.79 \times 10^{-2} \to 3.16 \times 10^{-2}$, while the phase speed converged cleanly at order 2; the amplitude history showed a beat. Refreshing $\boldsymbol{b}$ before stepping the fluid removed both. `MagneticSolver::step` takes a `field_is_current` flag so a coupled driver does not pay for the field pass twice.

A non-converging error sitting beside a converging one is the signature of a coupling-order problem, and any future two-way coupling should be checked the same way.

10 Validation

Every scheme is accompanied by the test that establishes it. The suite runs in both FP64 and FP32 and consists of ten executables; the numbers below are FP64 unless stated.

10.1 Poiseuille flow: wall placement and the magic parameter

Force-driven plane Poiseuille flow, periodic in x, halfway bounce-back walls in y. With H fluid nodes the no-slip planes sit at $y = 0.5$ and $y = H + 0.5$, so

$$u(y) = \frac{G}{2\rho\nu}\,(y - 0.5)\,(H + 0.5 - y), \qquad u_{\max} = \frac{GH^2}{8\rho\nu}. \tag{69}$$

Which relaxation rates play the roles in (30) depends on the operator, so each has its own magic point. Measured wall slip (D2Q9, $H = 32$):

τ	BGK	TRT(3/16)	MRT	central moments
0.6000000	-7.40×10^{-3}	$+2.51 \times 10^{-14}$	-5.73×10^{-3}	-5.73×10^{-3}
0.8000000	-4.06×10^{-3}	$+7.53 \times 10^{-14}$	-1.56×10^{-3}	-1.56×10^{-3}
0.8750000	-1.95×10^{-3}	-1.92×10^{-14}	$\mathbf{+4.11 \times 10^{-14}}$	$\mathbf{+7.39 \times 10^{-14}}$
0.9330127	$\mathbf{-1.08 \times 10^{-14}}$	$+5.35 \times 10^{-14}$	$+1.21 \times 10^{-3}$	$+1.21 \times 10^{-3}$
1.2000000	$+1.26 \times 10^{-2}$	$+3.55 \times 10^{-15}$	$+6.77 \times 10^{-3}$	$+6.77 \times 10^{-3}$

TRT holds the wall at 0.5 to $\sim 10^{-14}$ at *every* viscosity. BGK manages it only at $\tau = \frac{1}{2} + \frac{\sqrt{3}}{4} = 0.9330127$. The moment operators relax third-order modes at unity, so their magic point is $\left(\frac{1}{\tau} - \frac{1}{2}\right)\frac{1}{2} = \frac{3}{16}$, i.e. $\tau = 7/8$ exactly — *predicted before the run and confirmed to* 7×10^{-14}.

Resolution behaviour. At its magic point an operator is exact at every resolution. Away from it (FP64, D2Q9):

operator @ τ	L_2, H=16	H=32	H=64	ord(L_2)	ord(slip)
BGK @ 0.9330127	4.2e-14	1.2e-14	5.8e-13	exact	exact
TRT(3/16) @ 0.6	4.6e-14	1.1e-13	1.6e-13	exact	exact
MRT @ 0.875	3.2e-14	9.3e-14	3.1e-13	exact	exact
central moments @ 0.875	3.6e-14	8.4e-14	3.2e-13	exact	exact
BGK @ 0.6 (off-magic)	5.1e-03	1.3e-03	3.2e-04	**2.00**	**1.00**

Off-magic bounce-back is still *second* order: a lattice-aligned wall does not degrade the scheme. The pair of orders is what identifies the defect — the discrete error is a constant velocity offset $C \sim (\Lambda - 3/16)\,G/\nu$, which at fixed peak velocity scales as $1/H^2$, while the fitted wall *position* moves only as $1/H$ because the parabola's curvature shrinks with it. Seeing 2 and 1 together says "offset", not "displaced wall".

These two assertions are FP64-only by design: at $H = 64$ the discretisation error is 3×10^{-4}, which in FP32 lies below the accumulated round-off floor of about 10^{-3}. Loosening the tolerance would not fix that, it would only stop the test measuring anything.

10.2 Decaying flows with exact solutions

Three periodic flows, chosen so that each isolates something the others cannot.

Taylor–Green $\boldsymbol{u} = U(-\cos kx \sin ky,\ \sin kx \cos ky)\, e^{-2\nu k^2 t}$. Nonlinear and exact, but z-independent — so it is the *same 2D problem* on every lattice. It is a cross-lattice consistency check, not a 3D test.

ABC / Beltrami $\boldsymbol{u} = U(\sin kz + \cos ky,\ \sin kx + \cos kz,\ \sin ky + \cos kx)$. Here $\nabla \times \boldsymbol{u} = k\boldsymbol{u}$, so $\boldsymbol{u} \times (\nabla \times \boldsymbol{u}) = \mathbf{0}$ and the nonlinear term collapses to $\nabla(|\boldsymbol{u}|^2/2)$, absorbed by the pressure: $\boldsymbol{u}(t) = \boldsymbol{u}_0 e^{-\nu k^2 t}$ exactly. Every component varies in every direction. This is the nonlinear 3D convergence test.

Diagonal shear wave $\boldsymbol{u} = U\boldsymbol{e} \sin(\boldsymbol{K}\cdot\boldsymbol{x}) e^{-\nu|K|^2 t}$ with $\boldsymbol{K} = \frac{2\pi}{L}(1, 1, 1)$ and $\boldsymbol{e}\cdot\boldsymbol{K} = 0$. Linear by construction, and off-axis, so it probes lattice isotropy where an axis-aligned wave cannot.

The three are complementary and none substitutes for another. The ABC flow is built from axis-aligned sinusoids — it tests nonlinear 3D convergence but says nothing about isotropy.

Interpreting the effective viscosity. ν_{eff} does *not* equal ν at finite resolution: a sinusoid decays at the rate set by the discrete Laplacian, which differs from νk^2 at $O(k^2)$. What is asserted is therefore that the error converges away at second order, not that it is small at any given N. Diffusive scaling $U \sim 1/N$ keeps the Mach-number error second order alongside the spatial error.

10.3 Grid convergence: all lattices, all operators

A dedicated sweep (`validation/grid_convergence`) covers every lattice × operator combination over a resolution ladder and reports 92 data points. Slopes are least-squares fits of log(error) against $\log N$ over the whole ladder, not pairwise ratios.

case	lattice	operator	slope ν err	slope $L_2(u)$	worst R^2
Taylor–Green	D2Q9	BGK, TRT, MRT, CM	−2.02	−1.99	
Taylor–Green	D3Q27	BGK, TRT, MRT, CM	−2.02	−1.99	
ABC/Beltrami	D3Q27	BGK, TRT, MRT, CM	−2.01	−2.01	
24 fitted slopes, range −2.023 **to** −1.994					$R^2 \geq 0.99982$

No operator gains or loses an order. What differs is the error *amplitude*, and no operator dominates everywhere:

	ABC flow, ν error at $N = 48$	diagonal shear wave, $N = 64$
BGK	2.75×10^{-3}	1.00× (reference)
TRT(3/16)	$\mathbf{1.63 \times 10^{-3}}$	—
MRT	2.15×10^{-3}	—
central moments	2.15×10^{-3}	**1.38**× better (D3Q27)

TRT is the most accurate in a periodic bulk flow, because the magic parameter tunes the third-order modes that the moment operators send straight to equilibrium. On the off-axis wave the ordering reverses.

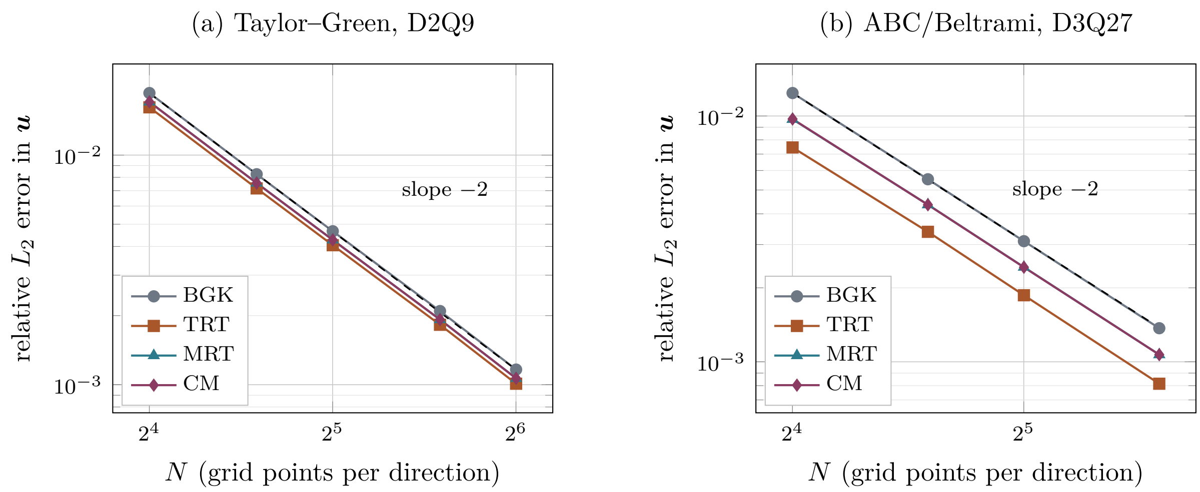


Figure 1: Grid convergence of the four collision operators, log–log, with a slope -2 reference (dashed). Diffusive scaling $U \sim 1/N$. Panel (a) is the Taylor–Green vortex on D2Q9; D3Q27 lies exactly on top of these points (spread 3.9×10^{-14}) because the field is z-independent. Panel (b) is the ABC/Beltrami flow on D3Q27. The lines are parallel: the operators differ in error *amplitude*, not in order. Fitted slopes over the whole ladder range from -1.994 to -2.023 across all 24 series.

10.4 Galilean invariance

Superimposing a uniform velocity U_0 on a solution only translates it, so the recovered viscosity must not depend on U_0. ABC/Beltrami flow, $N = 32$, $\tau = 0.8$:

lattice	operator	U_0=0	0.05	0.10	0.15	drift vs BGK
D3Q27	BGK	0.100620	0.099864	0.097595	0.093812	—
D3Q27	TRT(3/16)	0.100368	0.099618	0.097368	0.093617	1×
D3Q27	MRT (raw)	0.100484	0.100485	0.100486	0.100490	1153×
D3Q27	central moments	0.100484	0.100484	0.100482	0.100479	**1439×**

Two findings deserve emphasis.

1. **TRT is no better than BGK here.** The magic parameter fixes where the *wall* sits, not which frame the collision prefers. These are independent defects and it takes both operators to address both.
2. **Most of the gain is the complete equilibrium, not the co-moving frame.** Raw MRT already recovers 1153×, because this implementation gives it the full product-form equilibrium; centring the basis adds a further $\sim 25\%$. The split may differ for flows whose nonlinearity is less balanced than a Beltrami field, but the effect should not be attributed wholly to centring.

Every residual scales as U_0^2 (fitted powers 2.00–2.10), so what remains is the weakly-compressible $O(\mathrm{Ma}^2)$ error of the method itself, not frame dependence of the collision.

10.5 Thermal transport

Two exact solutions of the advection–diffusion equation, with amplitude and Peclet number both scaled diffusively ($U \sim 1/N$) so that the comparison is a convergence study rather than a measurement of how far the wave travelled:

case	order (FP64)	order (FP32)	D_{eff} rel. err	phase err
D2Q5 diffusion	2.00	2.03	1.5×10^{-3}	4.5×10^{-14}
D3Q7 diffusion	2.00	2.00	1.3×10^{-3}	2.8×10^{-14}
D2Q5 advection–diffusion	2.00	2.02	1.1×10^{-3}	2.5×10^{-4}
D3Q7 advection–diffusion	1.99	2.00	7.2×10^{-4}	2.3×10^{-4}

The phase is measured as well as the amplitude: an error in the coupling term puts the wave in the wrong *place*, which an amplitude check alone would not detect.

10.6 Natural convection

Differentially heated square cavity, hot and cold vertical walls, adiabatic horizontal walls, no-slip throughout, against the benchmark of de Vahl Davis [7] at Pr = 0.71. Fixing the characteristic velocity $u_c = \sqrt{g\beta\,\Delta T\,H}$ sets the Mach number, and then $\nu = H u_c \sqrt{\mathrm{Pr}/\mathrm{Ra}}$ follows.

Ra	N	op	Nu	benchmark	dev.	$u_{\max}$	reference
10^3	64	BGK	1.1176	1.118	−0.04%	3.65	3.649
		TRT	1.1179	1.118	−0.01%	3.65	
10^4	64	BGK	2.2431	2.243	+0.00%	16.14	16.178
		TRT	2.2479	2.243	+0.22%	16.14	
10^5	96	BGK	4.5169	4.519	−0.05%	34.47	34.730
		TRT	4.5323	4.519	+0.29%	34.52	

Within 0.3% everywhere. Note that BGK slightly edges TRT here, contrary to the expectation that the magic parameter would help: the wall placement it fixes is a *momentum* boundary, whereas what limits this case is the thermal boundary layer, resolved by the scalar lattice.

10.7 Rayleigh–Bénard convection

A layer of depth H heated from below and cooled from above, periodic horizontally, no-slip and isothermal on both plates. This exercises the scalar Dirichlet condition in the configuration it was designed for, and couples it to the buoyancy force of §7.3.

Wall pairing. The scalar uses anti-bounce-back and the fluid halfway bounce-back, so *both* wall planes sit at $y = 0.5$ and $y = H + 0.5$ and the layer is exactly H lattice units deep. Pairing an on-node condition for one field with a midway condition for the other would place the thermal and viscous boundaries half a spacing apart, which corrupts Ra itself rather than merely adding an error. The box is one critical wavelength wide, $2\pi/k_c = 2.0158\,H$.

Onset. Linear stability of a rigid–rigid layer gives $\mathrm{Ra}_c = 1707.762$ at $k_c H = 3.117$. This is the sharpest single number this configuration can be asked for: the buoyancy coupling, both boundary conditions, the diffusivity calibration and the wall placement all feed into it, and an error in any one moves it. It is measured here by bisecting on the sign of the growth rate of the critical mode.

	bounce-back + anti-bounce-back		regularised + moment	
H	Ra_c	deviation	Ra_c	deviation
16	1764.0	+3.29%	1769.5	+3.61%
24	1730.5	+1.33%	1737.4	+1.73%
32	1718.9	+0.65%	1725.7	+1.05%
48	1712.1	**+0.25**%	1716.8	+0.53%

Both families converge on 1707.762, the midway pair at $O(h^{2.3})$ and the on-node pair at $O(h^{1.6})$.

Measuring the growth rate. The obvious estimator — the total kinetic energy — does not work. Seeded at 10^{-6} it sits near round-off, and the resulting "growth rate" is not even monotonic in Ra, which is impossible for a linear instability; it put Ra_c at 1749 rather than 1719 at $H = 32$. Projecting the vertical velocity onto the critical mode $\sin(kx)\sin(\pi y)$ instead rejects acoustic transients, round-off and the spurious currents of the initial state, and recovers a growth rate that varies smoothly and crosses zero cleanly.

Nusselt number. At $H = 48$, $\mathrm{Pr} = 0.71$. Two independent estimators are reported — the wall gradient and the volume-averaged convective flux $1 + \langle vT\rangle H/(D\,\Delta T)$. They measure different things, so their agreement is a convergence check and not a tautology.

Ra	Ra/Ra_c	Nu (wall)	Nu (volume)	spread	$\lvert u\rvert_{\max}$
1 500	0.88	1.0000	1.0000	6.0×10^{-13}	0.0000
2 500	1.46	1.4692	1.4690	1.3×10^{-4}	0.0038
5 000	2.93	2.1079	2.1073	3.0×10^{-4}	0.0059
10^4	5.86	2.6490	2.6475	5.7×10^{-4}	0.0073
3×10^4	17.6	3.6144	3.6089	1.5×10^{-3}	0.0090
5×10^4	29.3	4.1523	4.1424	2.4×10^{-3}	0.0096
10^5	58.6	4.9869	4.9659	4.2×10^{-3}	0.0103

Below onset the result is exact: $Nu = 1.0000$ with the two estimators agreeing to 6×10^{-13} and $\lvert u\rvert_{\max}$ identically zero, so the conductive state is a fixed point of the coupled system to round-off. Above onset the estimators agree to better than 0.1% up to $\mathrm{Ra} = 10^4$ and to 0.42% at 10^5; the growth of that spread is the honest indicator that the top of the table is the least settled part. A power-law fit gives $Nu = 0.132\,\mathrm{Ra}^{0.319}$ overall and an exponent of 0.267 for $\mathrm{Ra} \geq 3 \times 10^4$.

What is not claimed. These Nu have *not* been compared against published tables. Values for two-dimensional rolls depend on aspect ratio and Prandtl number, and no reference set has been verified here, so no deviation column is given. What is verified is internal: the subcritical exactness, the agreement of two independent estimators, and Ra_c itself. Two further caveats apply to the highest rows: the box is one critical wavelength wide, which is the right choice for onset but restricts the roll structure at large Ra, and above $\mathrm{Ra} \sim 10^5$ two-dimensional convection may become time-dependent, in which case a single final-state Nu should be replaced by a time average.

10.8 Magnetohydrodynamics: exact solutions

case	order	recovered	rel. error
resistive decay, L_2	2.00	$\eta = 0.050110$	2.2×10^{-3}
shear Alfvén wave, phase speed	2.00	$v_A = 0.050017$	3.4×10^{-4}
shear Alfvén wave, damping	—	$(\nu + \eta)/2$	5.3×10^{-3}

The shear Alfvén wave is an exact solution of the *full nonlinear* equations, not merely the linearised ones: with $\boldsymbol{u} \perp \boldsymbol{b}_0$ and everything depending only on x, $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$ vanishes identically while $(\boldsymbol{b}\cdot\nabla)\boldsymbol{b}$ does not, so the Lorentz coupling and the induction equation are both driven and both must be correct. Phase is measured as well as amplitude, because an error in the coupling produces the wrong wave *speed* rather than the wrong amplitude.

On $\nabla\cdot\boldsymbol{b}$. The two wave cases report $|\nabla\cdot\boldsymbol{b}|$ at round-off, and *this is meaningless*: in both, b_x depends only on y and b_y only on x, so the divergence is structurally zero whatever the scheme does. Orszag–Tang is the only case in the suite that exercises the property, and there $\nabla\cdot\boldsymbol{b}$ is *not* preserved to machine precision: measured against the field's own gradient scale $k|\boldsymbol{b}|$ it sits at truncation level and converges away at order 1.65. That is the honest claim, and it is stated here in place of the stronger one.

10.9 Orszag–Tang vortex

The Orszag–Tang vortex [8] has no closed-form solution, so it is not validated against a formula. It is validated against published reference data and against structural properties. Initial conditions on a periodic box of side L:

$$\boldsymbol{u}(\boldsymbol{x},0) = u_0\big[-\sin y,\ \sin x\big], \qquad \boldsymbol{b}(\boldsymbol{x},0) = b_0\big[-\sin y,\ \sin 2x\big]. \tag{70}$$

10.9.1 Comparison with published data

Reproducing the setup of [6]: $L = 2\pi\,\mathrm{m}$, $u_0 = b_0 = 2$, $\rho = 1\,\mathrm{kg\,m^{-3}}$, $N = 1024$, $\Delta t = 5\times10^{-5}\,\mathrm{s}$. The derived lattice parameters reproduce the published values exactly ($u_{0,\mathrm{lat}} = 1.6297\times10^{-2}$, $\mathrm{Ma} = 0.0282$), with $\mathrm{Re} = u_0 N/\nu$ in lattice units and $\mathrm{Pr}_m = 1$. The reference column is the high-resolution pseudo-spectral data quoted there.

	t (s)	spectral	published LB	here, BGK	here, hybrid CM
$j_{\max}$	0.5	18.24	18.24	18.23	18.23
$j_{\max}$	1.0	46.59	46.65	46.55	46.55
$\zeta_{\max}$	0.5	6.758	6.756	6.756	**6.756**
$\zeta_{\max}$	1.0	14.20	14.18	14.182	**14.180**

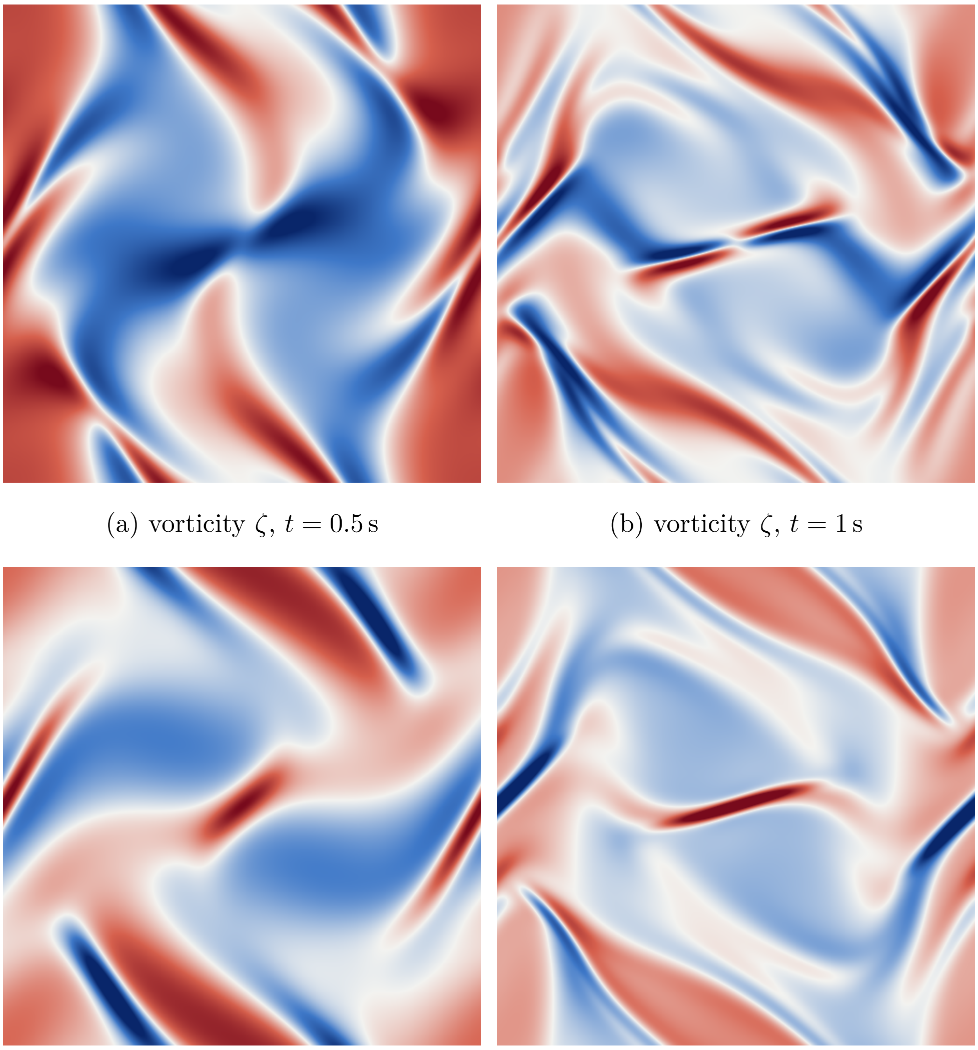
Errors against the spectral reference are 0.07/0.09/0.03/0.14 %, where the published LB errors are 0/0.13/0.03/0.14 %: the same accuracy from an independent implementation, with the hybrid scheme landing on the published vorticity values to every quoted digit. Convergence toward the reference is second order, $j_{\max}$ at $t = 1$ going -1.92% ($N = 256$), -0.46% ($N = 512$), -0.09% ($N = 1024$).

The one residual is $j_{\max}$ at $t = 1$, 0.22% below the published value. That is the quantity in which to expect it: the reference computes the current locally from the distributions whereas this code uses second-order central differences, and j is the more gradient-sensitive of the two peaks. Vorticity agreeing exactly while the current is 0.2% low points at the derivative operator rather than at the scheme.

10.9.2 Stability at high Reynolds number

At $\mathrm{Re} = 5000$ the lattice viscosity gives $\tau = 0.510$ on the 1024^2 grid, close enough to 1/2 that the BGK ghost modes go unstable. Runs are terminated at the first non-finite field or at $\mathrm{Ma} > 0.5$.

grid	τ	BGK	hybrid CM	published BGK
512^2	0.505	blow-up $t = 0.38\,\mathrm{s}$	stable to $t = 1.00\,\mathrm{s}$	—
1024^2	0.510	blow-up $\boldsymbol{t = 0.54}\,\mathrm{s}$	stable to $\boldsymbol{t = 1.20}\,\mathrm{s}$	$t \approx 0.52\,\mathrm{s}$



(a) vorticity ζ, $t = 0.5\,\mathrm{s}$ (b) vorticity ζ, $t = 1\,\mathrm{s}$

(c) current j, $t = 0.5\,\mathrm{s}$ (d) current j, $t = 1\,\mathrm{s}$

Figure 2: Orszag–Tang vortex at $\mathrm{Re} \approx 628$, $\mathrm{Pr}_m = 1$, on a 512^2 grid with the hybrid central-moment operator. Diverging colour scale, blue negative, red positive. The scale *saturates at the 99th percentile of* $|v|$ — 6.16 and 11.13 for ζ, 12.88 and 22.67 for j — because the current sheets are strongly localised: at $t = 1$ the peak $|j|$ is 46.4 against a 99th percentile of 22.7, so scaling by the maximum would leave the whole field pale. Between $t = 0.5$ and $t = 1$ the smooth initial vortex steepens into the thin diagonal current sheet visible in panel (d), which is the feature this benchmark exists to resolve and the reason $j_{\max}$ is the more grid-sensitive of the two reported peaks.

The BGK failure reproduces the published behaviour quantitatively — within one probe interval (0.02 s) — and shows the correct resolution dependence. Its signature is instructive: $j_{\max}$ grows smoothly to 24.0 at $t = 0.50$, triples to 73.7 at $t = 0.52$, then overflows, while the Mach number is 0.055 at the last clean sample. This is a collision instability, not a gradual compressibility breakdown.

The hybrid central-moment scheme runs 2.2× longer on the same grid and passes the $t \approx 0.99$ s mark that defeated BGK even on a finer grid in [6]. Its $j_{\max}$ peaks at 166.2 near $t = 1.00$ and then *decays* smoothly to 155.4 by $t = 1.18$, with the Mach number bounded at 0.069 throughout. That turnover is the clearest evidence the run is genuinely resolved rather than merely not-yet-diverged: an unstable run grows monotonically, it does not turn over.

10.9.3 Quasi-two-dimensional reduction: 3D lattices with $n_z = 1$

The Orszag–Tang vortex is a two-dimensional problem, so running it on a 3D lattice with a *single* cell in z is a reduction test: with $n_z = 1$ and periodic z the neighbour wrap sends the out-of-plane neighbour back to the node itself, the corresponding populations stream in place, and the answer must agree with the native 2D lattice. No code change was required — the wrap of §3.3 and the pair swap of Table 6 handle it.

The magnetic field follows the fluid's dimensionality by default (D2Q5 beside D2Q9, D3Q7 beside D3Q27), but since $b_z \equiv 0$ for this flow a 2D magnetic lattice is a legitimate pairing and isolates its contribution. Error against the spectral reference, per cent:

fluid	magnetic	$j_{\max}(0.5)$	$j_{\max}(1)$	$\zeta_{\max}(0.5)$	$\zeta_{\max}(1)$
$N = 256$					
D2Q9	D2Q5	−0.89	−1.92	−0.25	−1.57
D3Q27	**D2Q5**	**−0.89**	**−1.92**	**−0.25**	**−1.57**
D3Q27	D3Q7	−0.77	−1.66	−0.24	−1.51
$N = 512$					
D2Q9	D2Q5	−0.25	−0.46	−0.07	−0.48
D3Q27	D3Q7	−0.22	−0.39	−0.07	−0.46

Two results, one exact and one asymptotic.

D3Q27 reduces to D2Q9 exactly. Not approximately: the agreement in the highlighted row is structural, and both conditions for it can be checked in exact arithmetic. First, summing the D3Q27 weights over z reproduces the D2Q9 weights term for term,

$$\tfrac{8}{27} + 2 \cdot \tfrac{2}{27} = \tfrac{4}{9}, \qquad \tfrac{2}{27} + 2 \cdot \tfrac{1}{54} = \tfrac{1}{9}, \qquad \tfrac{1}{54} + 2 \cdot \tfrac{1}{216} = \tfrac{1}{36}, \tag{71}$$

so for a z-invariant field the projected equilibrium and the projected BGK update are exactly D2Q9's. Second — and this is the part that is not automatic — the out-of-plane Maxwell-stress contribution cancels within every in-plane group:

$$\sum_{c_z} w_i \left(c_z^2 - c_s^2\right) = 0 \qquad \text{for each } (c_x, c_y) \text{ on D3Q27}. \tag{72}$$

The spread converges away. Between $N = 256$ and $N = 512$ the lattice-to-lattice spread falls from 0.12, 0.26 and 0.06 percentage points (on $j_{\max}(0.5)$, $j_{\max}(1)$, $\zeta_{\max}(1)$) to 0.03, 0.07 and 0.02 — factors of 4.0, 3.7 and 3.0, i.e. second order. The lattice pairings are therefore converging to the same answer, and what separates them at finite resolution is truncation error rather than different physics. Of the two choices, the magnetic lattice matters more than the fluid one: swapping D2Q5 for D3Q7 moves $j_{\max}$ by about 0.25 percentage points at $N = 256$, which is unsurprising given $c_s^2 = 1/3$ against 1/4 and correspondingly different truncation in the induction equation.

10.10 Boundary conditions

Four tests, one per claim. All are second order, which is the accuracy of the scheme itself, so what distinguishes the conditions is *where the wall is*, and that is what each table measures rather than assumes.

Regularised walls: order and wall placement. Poiseuille flow with regularised walls on the boundary rows, $\tau = 0.8$. The wall position is obtained by fitting a parabola to the three interior nodes nearest the wall and taking its root; reading $\boldsymbol{u}$ back at the wall node would test nothing, since the solver reports the imposed value there by construction.

W	L_2 error	order	fitted wall position
8	1.708×10^{-2}	—	2.68×10^{-2}
16	4.419×10^{-3}	1.950	1.33×10^{-2}
32	1.123×10^{-3}	1.976	6.67×10^{-3}

The offset halves with each refinement: first order in lattice units, hence second order relative to the channel width. It is entirely the curvature slip (57): at $\tau = 0.8$, $\frac{4}{3}(\omega - 1)/\omega^2 = 0.2133$ against 0.2128 measured. At $\tau = 1$ the same run is exact to 1×10^{-15}.

These numbers are worse than they were before the force corrections of §8.1 were applied, and that is worth stating rather than hiding: without them the offset at $\tau = 0.8$ was 1.42×10^{-2}, because the force defect and the curvature slip carry opposite signs there and partially cancelled. Two errors cancelling is not accuracy, and removing the one that is understood is the right move even though a headline number rises. The order of convergence is unchanged. D2Q9 and D3Q27 agree to seven significant figures on this flow, the plane channel being quasi-one dimensional so that D3Q27 with periodic span reduces exactly.

Corners: the two closures compared. Lid-driven cavity, 129^2, against Ghia, Ghia and Shin [3].

Re	corner closure	$\max\lvert\Delta u\rvert$	$\max\lvert\Delta v\rvert$
100	local	0.0044	0.0071
100	finite diff.	0.0046	0.0065
1000	local	0.0187	0.0149
1000	finite diff.	**0.0123**	**0.0117**

At $Re = 100$ the two are indistinguishable — four corner nodes out of 16 641 do not move the centrelines. At $Re = 1000$ the finite-difference closure is better at *every* station of the u profile, by roughly a third in both the maximum and the rms. The improvement sits in the near-wall regions and vanishes in the core, which is the expected signature: the corner closure perturbs the boundary layers, and at $Re = 1000$ those are thin enough for the perturbation to reach the centreline. This is why the finite-difference route is the default.

Magnetic walls: Hartmann flow. The MHD analogue of Poiseuille flow, $\mathrm{Ha} = 10$, $\nu = \eta$, against the exact solution [2]

$$b(\xi) = \frac{FL}{B_0}\left[\frac{\sinh(H\xi)}{\sinh H} - \xi\right], \qquad u(\xi) = \frac{FL}{B_0}\sqrt{\frac{\eta}{\nu}}\coth H\left[1 - \frac{\cosh(H\xi)}{\cosh H}\right], \tag{73}$$

with $\xi = x/L$ and $H = B_0 L/\sqrt{\nu\eta}$. Velocity uses regularised walls, $\boldsymbol{b}$ uses the moment condition (63); both put the wall on the node, so they agree about where it is.

n	$\ell_2(u)$	order	$\ell_2(b)$	order	$\lvert u\rvert$ wall	$\lvert b\rvert$ wall
17	3.873×10^{-4}	—	4.117×10^{-4}	—	0	0
33	8.831×10^{-5}	2.133	1.009×10^{-4}	2.029	0	0
65	2.136×10^{-5}	2.047	2.544×10^{-5}	1.988	0	0
129	5.476×10^{-6}	1.964	6.572×10^{-6}	1.953	0	0

Second order in both fields, and both vanish at the walls exactly rather than to $O(h^2)$ — the property the moment condition exists to provide. The computed profile reproduces the flat Hartmann core, the $O(L/H)$ boundary layers, and $b(0) = 1.1 \times 10^{-17}$, antisymmetry at round-off.

Inlet-driven Hartmann flow: the same problem without a body force. The test above drives the channel with a uniform force through a streamwise-periodic domain. Removing the force and driving it from an inlet instead changes what is being tested in three ways, and each is the reason for doing it:

- It is the only way to put `MhdCentralMoments` against an analytic profile at all. That operator hardcodes `NoForcing` and cannot be body-force driven, so the force-driven test covers `MhdBGK` and could never have covered it.
- It is free of the Guo half-force interaction and the curvature slip (57), exactly as the inlet Poiseuille of §11.6 is.
- It is the only test in the suite exercising the fluid and magnetic outflow conditions at once.

Dellar's axes are rotated so the streamwise direction is x: walls on the two y-normal planes, half-width $L = (L_y - 1)/2$, $\xi = (y - L)/L$, and the applied field B_0 wall-normal along y, stretched by the flow into a streamwise b_x. The exact solution is (73) with x and y interchanged. The velocity inlet imposes the analytic u, the outlet is the constant back-pressure condition of §8.2, and $\boldsymbol{b}$ is imposed at the inlet but left free at the outlet — pinning it at both ends would tell the induction equation the answer at both boundaries and make the interior test partly circular.

L_y	rel $L_2(u)$	order	rel $L_2(b)$	order	$\lvert u\rvert$ wall	$\lvert b\rvert$ wall
17	2.281×10^{-2}	—	6.499×10^{-2}	—	0	0
33	5.391×10^{-3}	2.081	1.590×10^{-2}	2.031	0	0
65	1.286×10^{-3}	2.068	3.916×10^{-3}	2.022	0	0
129	3.071×10^{-4}	2.066	9.793×10^{-4}	2.000	0	0

$\mathrm{Ha} = 10$, $\nu = \eta = 0.1$, $u_{\max} = 0.005$, $L_x = 241$, $L_z = 1$. Second order in both fields, and both vanish at the wall node exactly. The Hartmann profile has a flat core with layers of thickness L/Ha, a tenth of the channel here, so this is a considerably harder test of the wall region than the parabolic Poiseuille profile.

Why L_x is so large. Not for physics: the magnetic outflow condition contaminates a long stretch upstream, as §8.4 records. With the $L_x = 21$ that the Poiseuille case uses, the contamination covers the entire domain and the measured order is meaningless. The Mach floor of §11.6 appears here too but is the smaller effect: at $u_{\max} = 0.02$ the last rate in u falls to 1.461, and dropping to $u_{\max} = 0.005$ recovers it.

Scalar walls: placement, measured. Steady one-dimensional conduction between walls at $T = 0$ and $T = 1$. The profile is exactly linear, so the residual is round-off ($\sim 10^{-14}$ throughout) and the only meaningful measurement is where a least-squares line through the *interior* nodes reaches the wall values.

	anti-bounce-back		moment condition	
N	$T = 0$ at	$T = 1$ at	$T = 0$ at	$T = 1$ at
9	0.500000	7.500000	−0.000000	8.000000
17	0.500000	15.500000	−0.000000	16.000000
33	0.500000	31.500000	−0.000000	32.000000
65	0.500000	63.500000	−0.000000	64.000000

Anti-bounce-back places its wall half a spacing outside the node, the moment condition exactly on it, at every resolution and to machine precision.

11 Validation campaign

The tests of §10 were written alongside the features they exercise. This section is different: it works through a published test suite without modification, so that the numbers can be put beside someone else's.

The suite is Sec. III of De Rosis and Coreixas [12], tests III.A to III.E, plus the inlet-driven Poiseuille flow of Sec. III.A of De Rosis [13]. Test III.F is out of scope: the campaign was specified to exclude applied body forces, and III.F has one.

Settings, and where they depart from the papers. Everything runs on the central-moments operator with the highest-order equilibrium each lattice admits (§5.8) — sixth order on D2Q9 and D3Q27. BGK and MRT were deliberately not run. The two-dimensional cases are run on D2Q9 and on D3Q27 with a single periodic point in z; III.E is genuinely three-dimensional and uses D3Q27. Fixed $u_0 = 0.02$ throughout rather than the papers' sweeps, and reduced resolutions where cost demanded: $D = 64$ rather than 128 in III.E, $D_{\text{lb}} = 512$ only in III.D, $N = 129$ rather than 201 in III.C. These are stated per test below because in two cases they change what the number means.

11.1 Taylor–Green vortex

Periodic square of side 2π, N^2 points, $\text{Re} = 1000$, error measured at $t = T$ against the exact decaying solution.

N	8	16	32	64	128	256
D2Q9	1.106×10^{-1}	2.566×10^{-2}	6.259×10^{-3}	1.444×10^{-3}	2.427×10^{-4}	6.010×10^{-5}
D3Q27	1.106×10^{-1}	2.566×10^{-2}	6.259×10^{-3}	1.444×10^{-3}	2.427×10^{-4}	6.010×10^{-5}

Fitted rate 2.186 on both lattices. The paper reports an optimal rate of 2 at its lowest u_0.

A correction to the published initial condition. Reference [12] prints $\boldsymbol{u}_0 = u_0[\cos\gamma x\ \sin\gamma y,\ \sin\gamma x\ \cos\gamma y,$ whose divergence is $-2u_0\gamma \sin\gamma x \sin\gamma y \neq 0$. Almost certainly a lost minus sign in typesetting; the standard solenoidal field, with $-\cos\gamma x \sin\gamma y$ in the first component, is used here. The printed density likewise carries a leading factor 3 that would give $\rho \approx 3$, and the standard form is used instead.

11.2 Double shear layer

$L = 256$, $\mathrm{Re} = 30\,000$, $\tau = 0.500512$ — a stability test rather than an accuracy one, run to $t/t_0 = 1$. All three lattices complete without divergence at that τ, and the normalised invariants agree closely:

	E/E_0	Ψ/Ψ_0
D2Q9	0.986436	0.622701
D3Q27	0.986436	0.622701

11.3 Lid-driven cavity

129^2, $u_{\mathrm{lid}} = 0.02$, against Ghia, Ghia and Shin [3]. Reported as the largest absolute deviation over the 17 tabulated stations, in lid-velocity units.

	Re = 100		Re = 400		Re = 1000	
	$\max\lvert\Delta u\rvert$	$\max\lvert\Delta v\rvert$	$\max\lvert\Delta u\rvert$	$\max\lvert\Delta v\rvert$	$\max\lvert\Delta u\rvert$	$\max\lvert\Delta v\rvert$
D2Q9	0.0049	0.0067	0.0069	0.0071*	0.0094	0.0099
D3Q27	0.0049	0.0067	0.0069	0.0071*	0.0094	0.0099

* *One published reference value is anomalous, and it is Ghia's, not a transcription error here.* All three tables in `validation/cavity_cm.cpp` have been checked entry-for-entry against an independent digitisation of [3] and they match. But the v entry at $\mathrm{Re} = 400$, $x = 0.9063$ reads -0.23827, and three things say it cannot be right:

- Ghia's own extremum table gives $\min v = -0.44993$ at $x = 0.8594$ for this Reynolds number. The three tabulated neighbours are then -0.44993 (0.8594), -0.23827 (0.9063), -0.22847 (0.9453): essentially the whole recovery from the minimum happens in one interval, after which the profile is flat.
- The same three stations are smooth at $\mathrm{Re} = 100$ (-0.22445, -0.16914, -0.10313) and at $\mathrm{Re} = 1000$ (-0.42665, -0.51550, -0.39188). Only $\mathrm{Re} = 400$ kinks, and at exactly one point.
- Every other station agrees here to 0.0071 or better; this one disagrees by 0.1415, a factor of twenty. The value computed here, -0.37974, is what a smooth profile through Ghia's own minimum would give.

The entry is left in the table — it is what the paper prints, and quietly substituting a number nobody published would be worse — but it is excluded from the headline score above and reported separately by the run. It could not be checked against the original paper, which is not among the references to hand. An earlier draft of this document blamed the discrepancy on a faulty transcription on this side; that was wrong, and the correction is recorded here rather than silently applied.

11.4 Dipole–wall collision

$D_{\mathrm{lb}} = 512$ (a 1025^2 grid), walls regularised, corners on the finite-difference closure — the only test in the campaign where the corner treatment matters, since III.A, III.B and III.E are periodic. Against Table II of [12], which reproduces the lattice Boltzmann results of Mohammed, Graham and Reis [14] and the spectral reference of Clercx and Bruneau [15]:

Re	t	energy E here	[14]	[15]	enstrophy Ψ here	[14]	[15]
625	0.25	1.4976	1.494	1.501	466.3	467.2	472.1
	0.5	1.0167	1.010	1.013	387.2	374.0	382.6
	0.75	0.7702	0.765	0.767	251.3	244.8	256.0
2500	0.25	1.8355	1.838	1.848	689.6	705.3	725.6
	0.5	1.5379	1.534	1.540	889.1	898.1	917.6
	0.75	1.3274	1.320	1.325	790.3	790.2	809.9

Energy agrees to between 0.1% and 0.7% everywhere. Enstrophy is the more demanding invariant and the agreement is between 0.01% and 3.5%; in most rows the value computed here falls *between* the published lattice Boltzmann result and the spectral one, and at Re $= 2500$, $t = 0.75$ it matches the published lattice Boltzmann enstrophy to four significant figures (790.3 against 790.2). D2Q9 and D3Q27 are identical to every digit shown.

An error of ours, and how it was caught. The published initial condition is written in physical units with $|\omega_e| = 299.56$, which makes $E(0) = 2$ and $\Psi(0) = 800$ exactly. It was first scaled so that the *peak* speed was u_0, which is wrong: the Reynolds number is built on the rms velocity, and for this field the peak-to-rms ratio is 11.019. The nominal Reynolds number and the elapsed physical time were both a factor eleven out, the flow dissipated far too slowly, and the discrepancy grew with t — a signature that pointed at the time scale rather than at the wall. Verifying Eq. (32) numerically ($E(0) = 2.0004$, $\Psi(0) = 799.96$, $U_{\mathrm{rms}} = 1.0001$) settled it. The results above are from the corrected normalisation.

11.5 Three-dimensional Taylor–Green vortex

$D = 64$, run to $t^* = tu_0/D = 10$. There is no exact solution; the comparison is between lattices. At the campaign's fixed $u_0 = 0.02$ the Mach number is 0.035 rather than the paper's 0.2–0.6, so a unit of physical time costs up to seventeen times more steps than in the paper.

	Re $= 1600$	Re $= 30\,000$
D3Q27	completes, $E/E_0 = 0.0145$ at $t^* = 10$	completes, $E/E_0 = 0.0399$ at $t^* = 10$

At Re $= 30\,000$ and this Mach number $\tau = 0.50026$. That case was initially judged infeasible on the available hardware; the judgement was wrong, and D3Q27 runs it to $t^* = 10$ without difficulty.

11.6 Inlet-driven Poiseuille flow

Sec. III.A of [13]. Laminar flow along x; no-slip on the two y-normal planes; the parabolic profile $u_x(0, y) = 4U_0\,(y/L_y)(1 - y/L_y)$ with $U_0 = 0.005$ imposed at the west face; outflow opposite; $L_x = 21$; Re $= U_0L_y/\nu = 10$; started from rest and run to steady state. The error is the relative L_2 norm, Eq. (15) there, between the analytic profile — which equals the inlet profile, the flow being fully developed — and a y-aligned cross section at mid-domain. $L_z = 1$ rather than the paper's 21; the flow has no z dependence and z is periodic either way. The outflow is the constant back-pressure condition of §8.2.

This test is worth having precisely because it is driven by the inlet and not by a body force, so it is free of both the Guo half-force interaction and the curvature slip (57) that dominate the forced channel of §10.1.

L_y	41	81	161	321	641	fitted rate
D2Q9	1.160×10^{-3}	3.251×10^{-4}	7.261×10^{-5}	1.558×10^{-5}	6.192×10^{-6}	1.964
D3Q27	1.160×10^{-3}	3.251×10^{-4}	7.261×10^{-5}	1.558×10^{-5}	6.192×10^{-6}	1.964

Against the published 2.063 that is 4.8% low. The shortfall is entirely in the last refinement, where the pairwise rate drops from 2.231 to 1.334, and it is understood — see below. It is worth noting first that the paper states its error at the finest grid, $L_y = 1281$, is of order 10^{-5}; the error here at $L_y = 641$ is 6.2×10^{-6}, already below that.

What limits the order — three falsified hypotheses. The outlet was the obvious suspect, and it is not the cause.

1. *Per-station error.* Breaking the error down by x at $L_y = 641$ makes the outlet look guilty: against $L_y = 161$ the inlet station converges at 1.99 and the bulk minimum at 1.92, both clean second order, while the outlet station converges at 0.70.
2. *Moving the outlet away does not help.* Repeating at $L_x = 41$ and $L_x = 81$ makes the last pairwise rate *worse*, not better: $1.334 \to 0.885 \to 0.007$. At $L_x = 81$ the error stops responding to refinement altogether — 5.974×10^{-6} at $L_y = 321$ against 5.945×10^{-6} at $L_y = 641$.
3. *A second-order outlet does not help.* Replacing the zero-gradient tangential extrapolation by a linear one (§8.2) is worse on the coarse grids and indistinguishable on the finest.

What it actually is: a non-refining, velocity-dependent error. Sweeping U_0 at fixed Re = 10 moves the onset of the degradation, and always toward *coarser* grids as U_0 rises.

U_0	Ma	fitted rate	pairwise rates			
0.0025	0.0043	**2.067**	1.720	2.026	2.205	2.263
0.005	0.0087	1.964	1.868	2.182	2.231	1.334
0.01	0.0173	1.523	2.091	2.179	0.974	0.812
0.02	0.0346	0.844	2.097	0.150	0.644	0.947

At $U_0 = 0.0025$ the velocity-dependent term stays below the discretisation error across the whole range and the fitted rate is 2.067, against the published 2.063 — agreement to 0.2%. The scheme is second order; the shortfall at $U_0 = 0.005$ is an artefact of the test's Mach number, not a defect of the wall or the outlet.

This is not a novel observation. Reference [12] reports the same behaviour on its own Taylor–Green convergence study — "for the lowest value of u_0, an optimal convergence value equal to 2 is found. As u_0 increases, the accuracy and convergence properties of the method deteriorate as well" — and attributes it to the cubic aliasing defect $c_{i\zeta}^3 = c_{i\zeta}$, which prevents the diagonal terms u_ζ^3 from entering the third-order equilibrium moments and can only be removed by explicit correction terms. The paper on this Poiseuille case [13] instead asserts that $U_0 = 0.005$ gives a Mach number "sufficiently low to annihilate detrimental compressibility effects"; at the accuracy reached here that assertion does not hold, which is unsurprising given that the error here at $L_y = 641$ is already below the error there at $L_y = 1281$.

Fitting $\varepsilon = A/L_y^2 + B$ to the two finest grids gives $B < 0$ at $U_0 = 0.0025$ — no floor reached — and $B = 3.1 \times 10^{-6}$, 9.2×10^{-6}, 2.1×10^{-5} at $U_0 = 0.005$, 0.01, 0.02. That is growth somewhere between linear and quadratic in U_0, but a two-point extraction is far too noisy to pin the exponent and none is claimed.

11.7 Orszag–Tang vortex in three dimensions

Section III.D and Fig. 6 of [13], with the initial condition from Mininni, Pouquet and Montgomery [11], that paper's Ref. [38]. Cubic periodic box of side 2π on M^3 points, $\delta = 2\pi/M$:

$$\boldsymbol{u}(\boldsymbol{x}, 0) = v_0 \left[-2\sin(\delta y),\ 2\sin(\delta x),\ 0\right], \tag{74}$$

$$\boldsymbol{b}(\boldsymbol{x}, 0) = 0.8\, b_0 \left[-2\sin(2\delta y) + \sin(\delta z),\ 2\sin(\delta x) + \sin(\delta z),\ \sin(\delta x) + \sin(\delta y)\right]. \tag{75}$$

Both are solenoidal: each component depends only on coordinates it is not differentiated by, so every term of the divergence vanishes identically. $\mathrm{Re} = 100$ on the lattice definition $\mathrm{Re} = v_0 M/\nu$, $\mathrm{Pr}_m = 1$, D3Q27 for the fluid and D3Q7 for the field, central moments throughout.

Three things the paper leaves open. Stated here because they change the numbers, not buried in the code.

- *A sign.* Equation (28) as printed has $+2\sin(\delta y)$ in the first velocity component; Ref. [11] has $-2\sin y$, which is what is used here. Unlike the two-dimensional Taylor–Green case in the same paper, the divergence does *not* discriminate — both forms are solenoidal — so this rests on the cited source rather than on a consistency check. It is a genuine ambiguity, not a demonstrated typo.
- v_0. The paper says only that "the values of v_0 and b_0 lead to a Mach number, $\mathrm{Ma} \sim 0.034$", and that depends on which speed Ma is built on. Taken here on the peak initial speed $2\sqrt{2}\, v_0$, giving $v_0 = 6.94 \times 10^{-3}$.
- b_0. Not given; set to v_0, the usual equipartition choice.

What can be checked. Figure 6 is a plot against a high-resolution pseudospectral run and those values are not available numerically, so no per-point comparison is possible. What the figure *does* supply is its axes — panel (a) plots the volume-averaged kinetic energy Γ normalised from 1, panel (b) runs from 5 to 25 in $J_{\max}$ — and the text states that "the current shows a maximum at about $t = 1$".

M	$J_{\max}$ peak	at t	$\Gamma(4)$
32	15.71	1.194	0.1240
64	19.19	1.198	0.1356
128	20.43	1.200	0.1374

Every check the figure permits comes out right. The peak time converges to $t = 1.200$ against the stated $t \approx 1$; $J_{\max}$ reaches 20.4 on the finest grid, inside the 5–25 axis; Γ falls from 1 to 0.137, inside the 0–1 axis; and the curves converge monotonically upward with M, the finer grids resolving the current sheet better, which is the trend the figure reports.

Convergence, and what it does and does not establish. Equation (30) of [13] defines χ as the L_∞ norm of the relative discrepancy against the spectral reference, quoted there as of order 10^{-2} for the finest grid. With no reference values to hand, the same measure is computed against $M = 128$ as a stand-in:

against $M = 128$	Γ	$J_{\max}$
$M = 32$	1.56×10^{-2}	2.31×10^{-1}
$M = 64$	1.9×10^{-3}	6.05×10^{-2}

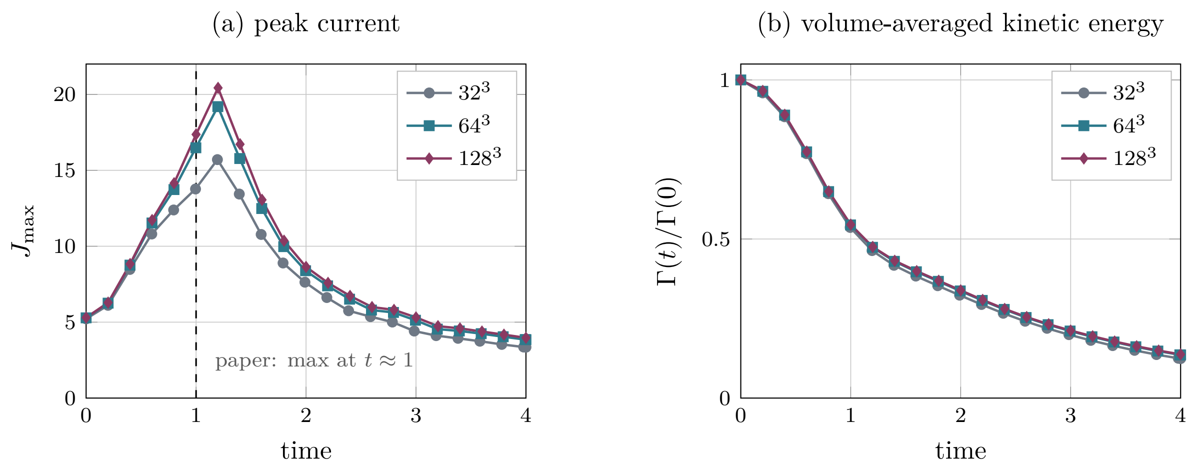


Figure 3: Three-dimensional Orszag–Tang vortex, D3Q27 with central moments, to be read against Fig. 6 of [13]. (a) The peak current magnitude rises with resolution as the current sheet is resolved, and its maximum converges to $t = 1.200$ against the paper's stated $t \approx 1$; the value on the finest grid, 20.4, sits inside that figure's 5–25 axis. (b) The energy, normalised as the paper normalises it, falls from 1 to 0.137. Both panels use a linear time axis; the published figure uses a logarithmic one.

The successive differences in the $J_{\max}$ peak are 0.171 and 0.061, a ratio of 2.83 — close to second order on a quantity that is a derivative of the field and therefore the hardest thing here to converge. The $M = 64$ figures, 6.1×10^{-2} on $J_{\max}$ and 1.9×10^{-3} on Γ, bracket the 10^{-2} the paper quotes.

This is self-convergence, not agreement. Substituting our own finest grid for the spectral reference flatters systematically, because it cannot see any error common to all three grids. It establishes that the scheme converges and roughly how fast; it does not independently establish that it converges to the right answer. Read the table above as a consistency check, not as a score.

One incidental result worth recording: at $M = 32$ this runs at $\tau = 0.507$, close enough to 1/2 that BGK ghost modes are normally a problem, and it completes the full $t \in [0, 4]$ without difficulty — consistent with the stability advantage [13] claims for central moments.

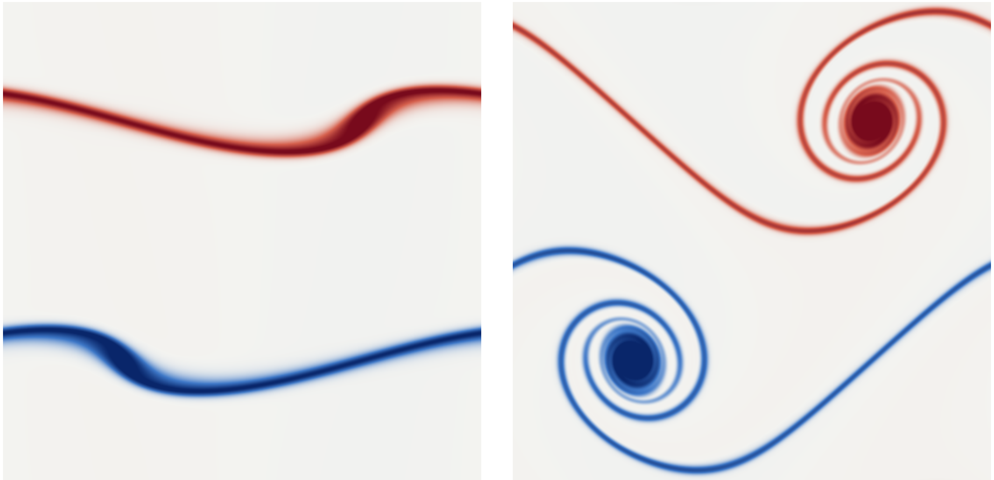
11.8 A finding that cuts across the campaign

D2Q9 and D3Q27 with one point in z are bit-identical. Not merely close: identical to every digit printed, in the Taylor–Green convergence, the shear layer, the dipole, the cavity and the inlet Poiseuille — five independent tests. The reduction of D3Q27 to D2Q9 under a single periodic z point is exact, which is a useful correctness check in both directions: it exercises the 3D code path against the 2D one on every one of these flows.

11.9 Field snapshots

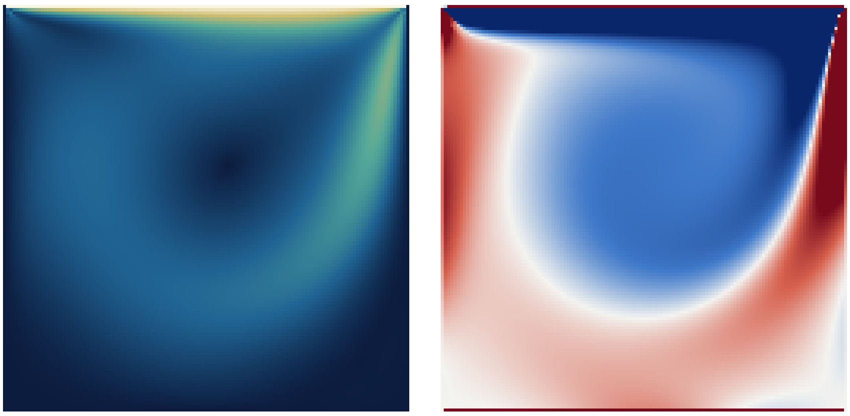
The tables elsewhere in this section say whether a number is right. These say whether the *flow* is right, which is a different question and one a norm cannot answer: a scheme can hit an integral quantity while getting the structure wrong. Every panel below is the field the corresponding test actually converged to, dumped from the run that produced the numbers quoted for it.

Signed quantities — vorticity, current — use a diverging map centred on zero; quantities with no meaningful zero — temperature, speed — use a sequential one. Where the colour saturates below the peak, the caption says so: these fields are strongly localised, and scaling by the maximum leaves the structure that matters invisible.



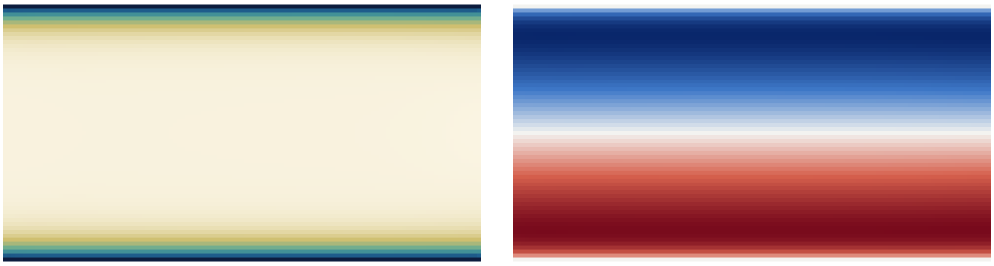
(a) $t/t_0 = 0.5$ (b) $t/t_0 = 1$

Figure 4: Double shear layer, vorticity, $L = 512$, $\mathrm{Re} = 30\,000$, D2Q9 central moments (§11). By (a) the perturbation has begun to distort the two layers; by (b) each has rolled into a vortex, and the two are joined by a single unbroken braid. That the braid stays thin and connected is the whole test: at $\tau = 0.50102$ the flow is barely resolved, and an under-dissipative or unstable scheme shows it here first, by fragmenting the braid into secondary vortices that are numerical rather than physical. Doubling the resolution from the $L = 256$ of the campaign table sharpens the structure and, more usefully, quantifies the numerical dissipation: the enstrophy retained at $t/t_0 = 1$ rises from 0.6227 to 0.6432, and the energy from 0.98644 to 0.98696. The physical dissipation at this Reynolds number is negligible over one turnover, so nearly all of that difference is the scheme's own, and it shrinks with resolution as it should. Colour saturates at the 99th percentile of $|\zeta|$.



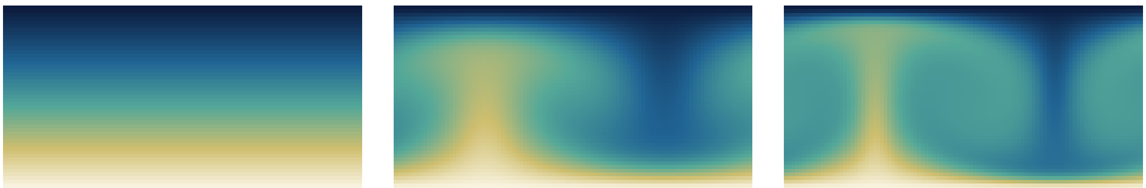
(a) speed $|\boldsymbol{u}|/u_{\mathrm{lid}}$ (b) vorticity

Figure 5: Lid-driven cavity at $\mathrm{Re} = 400$, 129^2, D2Q9 central moments, the run scored against Ghia *et al.* in §11 ($\max|\Delta u| = 0.0069$). The primary vortex sits above and right of centre and both downstream corner eddies are resolved. The vorticity is clipped hard, at the 88th percentile against a peak twenty times larger, because it is concentrated in the lid boundary layer; without that the interior is blank.



(a) velocity u_x (b) induced field b_x

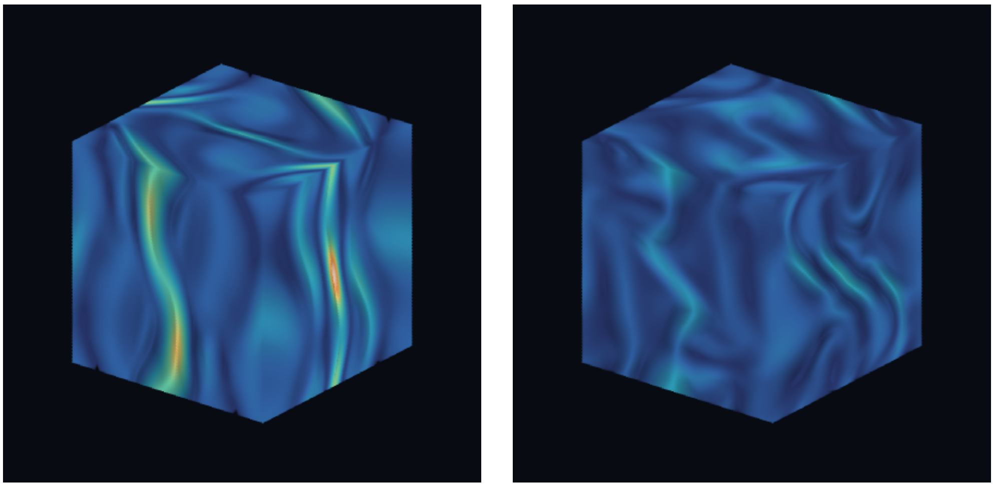
Figure 6: Inlet-driven Hartmann flow, $\mathrm{Ha} = 10$, $L_y = 65$, D2Q9 fluid with D2Q5 induction, central moments (§10.10). The velocity shows the flat Hartmann core with boundary layers of thickness L/Ha; the induced field is antisymmetric about the centreline, vanishing there and at both walls. Both are visibly independent of the streamwise coordinate, which is the property the exact solution has and which the convergence table depends on — any x structure here would be boundary error, and it is how the magnetic outflow defect (§8.4) was found.



(a) $\mathrm{Ra} = 1500$ (b) $\mathrm{Ra} = 10^4$ (c) $\mathrm{Ra} = 10^5$

$Nu = 1.0000$ $Nu = 2.6490$ $Nu = 4.9869$

Figure 7: Rayleigh–Bénard convection, temperature, $H = 48$, one critical wavelength wide, D2Q9 fluid with a D2Q5 scalar (§10.7). The three panels are the same run at three Rayleigh numbers and show what the Nusselt column of that section measures. (a) is *below* the critical $\mathrm{Ra}_c = 1707.762$: the perturbation decays and the field relaxes to pure conduction — a linear vertical gradient, no horizontal structure, and $Nu = 1$ to 6×10^{-13}. (b) is supercritical, and the same perturbation grows into a steady convection cell with a hot plume rising against cold fluid descending. By (c), at $58.6\,\mathrm{Ra}_c$, the thermal boundary layers at the two plates have thinned and the core is nearly isothermal — which is the mechanism behind Nu rising to 4.99, since Nu is precisely the wall gradient normalised by the conductive one. All three share a colour scale.



(a) $t = 1$, peak current

(b) $t = 2$

Figure 8: Three-dimensional Orszag–Tang vortex, current magnitude $|\nabla \times \boldsymbol{b}|$ through the whole 128^3 box, D3Q27 with D3Q7 induction (§11.7). Volume rendered rather than sliced: a cut plane shows the sheets only where they happen to intersect it and says nothing about how they are oriented, which for this flow is the point. The current organises into curved sheets wrapping the box, and (a) is the instant the peak is reached — the same $t \approx 1$ the $J_{\max}$ curve of Fig. 3 locates. *Both panels share one colour scale*, set by the $t = 1$ peak, so the visible dimming from (a) to (b) is the real decay: $J_{\max}$ falls from 19.1 to 8.32, less than half, in the same units as Fig. 3. Normalising each panel to its own maximum — the default in most rendering tools — would have made the later, weaker field the brighter of the two. The reference is recomputed at this resolution rather than carried over from the 64^3 volumes: the raw lattice values differ between the two by the factor $\Delta t \propto 1/M$ alone, so reusing the coarse-grid number would have rescaled the pair by about two. Orthographic view, azimuth 38°, elevation 24°; emission–absorption compositing with the weak field left transparent, the local gradient used as a surface normal for a diffuse term so the sheets read as surfaces rather than fog, and 2× supersampling. Rendered by `doc/fig/vol3d.py`, which is self-contained.

12 Demonstrator: flow in a patient-specific aorta

Everything else in this document runs on a geometry described by a formula. This one loads a voxel array off disk. It is included as a *demonstrator*, not as a validation case, and the distinction is worth stating precisely because the figures are seductive: there is no analytic solution here, no reference data is compared against, and **no number in this section is a claim of accuracy**. What it demonstrates is that the solver runs unmodified on a geometry with no symmetry and no closed form — `set_geometry` already takes an arbitrary predicate, so real geometry needed a reader, not a new solver.

12.1 Geometry and its verification

`src/io/VoxelGeometry.hpp` reads the flat binary produced by voxelising a SimVascular surface mesh (case `0074_H_AO_H`): $109 \times 184 \times 361 = 7\,240\,216$ voxels at a pitch of 0.0616 cm, tagged 0 solid, 1 fluid, 2 inlet, 3 outlet. Solid is 83.65% of the box; the lumen is 16.31%. The optional second inlet normal is detected by *file size*, not by stream state — guessing from the stream would silently read 24 bytes of the tag array as doubles on a file that does not carry one.

Three checks were run on the geometry before any flow was believed.

1. **Connectivity.** A six-connected flood fill seeded from an inlet voxel reaches 1 184 057 of 1 184 057 non-solid voxels: one lumen, no isolated pockets. This check exists because a single constant-x plane through this vessel cuts it into *two* disconnected regions — the aortic root and the arch — which looks exactly like a hole in the mesh and is not one. The projected silhouette is a single connected region, as the flood fill requires it to be.

2. **Caps.** Under six-connectivity the inlet fragments into 303 pieces and the outlets into 270. This is not damage: an oblique plane on a voxel grid is diagonally connected, not face-connected. By centroid they are one inlet and four outlets — three arch branches and the descending aorta.

3. **Domain edges.** 1283 fluid voxels sit on the box boundary carrying no cap tag. Were the domain periodic these would wrap, connecting the descending aorta to the arch and corrupting the field silently. It is not: the case constructs `Domain(nx,ny,nz,false,false,false)`, and independently there are *zero* fluid-to-fluid wrap pairs on any axis. The mesh is clipped flush against the box there and those voxels act as wall.

12.2 Boundary conditions

The vessel wall is halfway bounce-back on solid voxels, which under Esoteric Pull costs nothing: bounce-back is the identity on the storage, so solid cells are skipped outright — no load, no store, no arithmetic (§3.2). The population a fluid node emits toward the wall sits untouched for one step and is read back reversed on the next, placing the no-slip surface midway between the last fluid node and the first solid node. Halo cells are flagged `Excluded` and skipped by the same mechanism, so the box faces behave identically.

The inlet is a regularised velocity wall along the cap normal stored in the file. The cap is oblique to the voxel axes, so it has no single face normal; it is given `NrmCorner`, whose unknown set is built geometrically per node rather than from an axis. Using a face normal instead would drive flow into the wall.

The outlets are left at fixed pressure rather than a prescribed flow split, so the division of flow between the branch vessels is an *output* rather than an input. Imposing $\boldsymbol{u} = 0$ there would close the domain — measured as mass rising monotonically, $+8.5 \times 10^{-3}$ in 2000 steps.

Conserving outflow. The outlet overwrites its nodes every step, so whatever density is imposed is injected regardless of what arrived: pinning $\rho = 1$ makes the boundary a mass source and sink, and inlet and outlet fluxes then differ by $\sim$60% with no way to tell which is wrong. Closing a loop on *total mass* instead makes it conserving — raise the outlet density when mass is lost, lower it when mass accumulates. The fixed point is the density at which what leaves equals what enters, and it is found rather than assumed. Two details are load-bearing. The error must be normalised by total mass, not by outlet-node count: normalising by the latter gives an error of order 3, which saturates the clamp, reverses the flow and diverges the run in 1500 steps. And the controller is held off until the vessel has filled, since during filling mass *should* be changing and the loop otherwise fights the physics.

Note that the finite-difference corner-stress path is disabled for this case. It walks a two-node stencil off each wall node; in a box those neighbours are fluid or wall, but in a vessel they are frequently *solid*, whose populations Esoteric Pull never updates, so the stencil reads stale memory and feeds garbage stress into the inlet. The local closure is used instead until that path is made geometry-aware.

12.3 Steady inflow

$\mathrm{Re} = 50$ on the inlet cap's equivalent diameter $D = 2\sqrt{A/\pi} = 32.9$ lattice units, $U = 0.02$, $\nu = 1.3167 \times 10^{-2}$, $\tau = 0.5395$, D3Q27 central moments. After 13 500 steps the run is steady: $|\boldsymbol{u}|_{\max}$ flat at 7.2×10^{-2}, outlet density settled at 0.9934, and the mass error oscillating about zero at $\sim 1 \times 10^{-5}$ — which is the conserving outflow working.

The inlet flux is 12.98 in lattice units, and this is worth one sentence because measuring it is less obvious than it looks. Summing over *every* exposed face of the cap gives $|\sum \boldsymbol{n}_{\text{face}}| = 653.2$ pointing within 0.65° of the normal stored in the file, so $UA\cos\theta = 0.02 \times 653.2 \times 0.9935 = 12.98$. Filtering the faces by direction to exclude the vessel wall was tried and is *wrong* — it double-weights by the projection and drove the figure down to 10.38. The lateral wall faces already cancel in the vector sum; that is what makes the identity $\sum \boldsymbol{n}_{\text{face}} = A\boldsymbol{n}$ hold.

The outlet flux is **not** a valid measure for this boundary and is not used as one: the condition overwrites populations, so a flux summed through it reports the imposed state rather than what the flow delivered. Mass conservation is the trustworthy check.

12.4 Pulsatile inflow

The inlet is driven by the waveform used in the source project, reproduced unchanged so the two are comparable:

$$\frac{U(t)}{U_0} = \min\left(1, \frac{t}{t_{\text{ramp}}}\right)\Big[0.15 + 0.85\, w(\phi)^{3/2}\Big], \qquad w(\phi) = \tfrac{1}{2} + \tfrac{1}{2}\sin\!\Big(2\pi\phi - \tfrac{\pi}{2}\Big), \tag{76}$$

with $\phi = (t \bmod T)/T$: a ramp from rest, then a cycle between a diastolic floor at 15% of peak and a systolic peak, the upstroke sharpened by the 3/2 power. $\phi = 0$ is the diastolic minimum and the systolic peak falls at $\phi = 0.5$. Here $T = 2000$, $t_{\text{ramp}} = 800$, and U_0 is the systolic peak.

Since the direction is fixed by the cap normal and only the magnitude varies, the whole drive is a scale on the deduplicated wall-velocity table (`set_wall_velocity_scale`), not a per-node update, and costs nothing per step. The scale multiplies a saved copy of the original table rather than the live one: scaling in place would compound, and a waveform bounded in $[0, 1]$ would become a product of its own history and decay to zero within a few hundred steps — which would look like a plausible physical relaxation and is not one.

The parameter that characterises a pulsatile flow is not Re but the Womersley number, which fixes how far the oscillating shear layer penetrates in a beat:

$$\alpha = \frac{D}{2}\sqrt{\frac{2\pi}{T\nu}} = 8.04, \tag{77}$$

in the physiological range for an aorta. The drive is exact beat after beat: Q_{in} reaches -12.98043 at every systolic peak and -1.947065 at every trough, identical to five digits across six cycles.

Convergence. The viscous diffusion time across the vessel is $R^2/\nu = 20\,552$ steps, which is 10.3 beats. This run is 60 000 steps, 2.92 diffusion times, and it is converged: sampled at matched phase over the final three beats, *every* statistic of the speed field repeats to eight significant figures.

ϕ	step	mean	median	p_{99}	$\lvert\boldsymbol{u}\rvert_{\max}$
0.0	54000	0.00364663	0.00238384	0.0153191	0.0565515
0.0	56000	0.00364663	0.00238384	0.0153191	0.0565515
0.0	58000	0.00364663	0.00238384	0.0153191	0.0565515
0.5	55000	0.00483408	0.00495437	0.0104059	0.0320153
0.5	57000	0.00483408	0.00495437	0.0104059	0.0320153
0.5	59000	0.00483408	0.00495437	0.0104059	0.0320153

The residual differences are in the eighth digit, which is round-off. The outlet density has stopped moving at 0.99699, so the mass controller has found its fixed point rather than still walking toward one. An earlier 12 800-step run — 0.62 diffusion times — was reported in a previous draft of this section as *not* converged, with mean and median still rising monotonically through six beats. That was accurate, and it is what a run of that length looks like; it is superseded rather than corrected.

Phase relationships, and a storage effect that is not Womersley. Over a converged beat the *mean* speed peaks at $\phi = 0.60$ against a drive peaking at $\phi = 0.50$: a lag of 0.10 of a cycle, 36°, the expected sense and order for $\alpha = 8$.

The fast tail does the opposite. Both p_{99} and $\lvert\boldsymbol{u}\rvert_{\max}$ peak at $\phi = 0$ and reach their *minimum* at systole — nearly antiphase with the drive. This is not a transient: it repeats identically across the final three beats, and it is visible in Fig. 10, where at the diastolic trough the root has gone pale while the descending aorta carries the fastest flow in the vessel.

The mechanism is storage, not phase lag. Total mass is not constant over a beat: it swings by $\pm 2.7 \times 10^{-3}$, with its minimum at $\phi = 0.25$ and its maximum at $\phi = 0.70$, accumulating through systole and discharging through diastole. The fast tail peaks where that discharge is steepest. A transit-time explanation was considered and does not survive arithmetic — at these speeds a fluid particle needs several beats to cross the vessel.

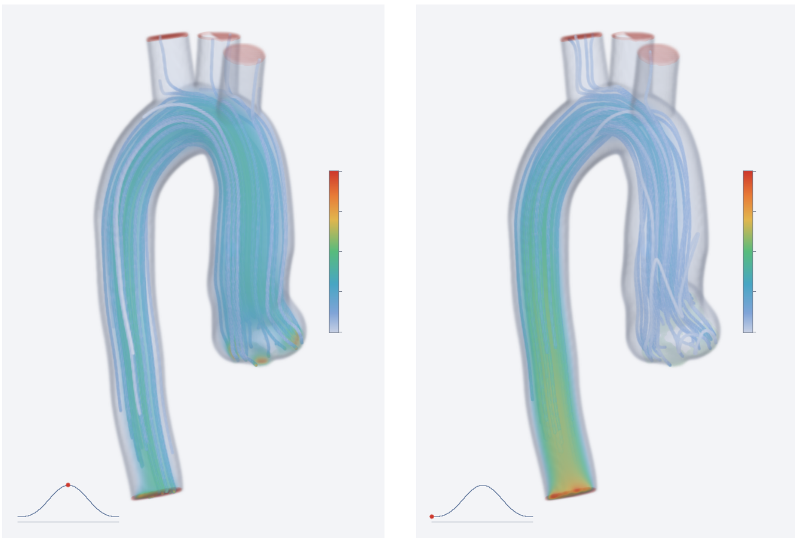
What this data does *not* settle is where the storage lives. Two candidates produce the same signature and are not distinguished by the dumps taken here, which carry velocity but not density: the lattice fluid's own $O(\mathrm{Ma}^2)$ compressibility acting as a compliance chamber, or the fixed-pressure outlet acting as a reservoir. Either way the effect is numerical rather than physiological — the wall is rigid (§12, and the Known Limitations) — so the antiphase should not be quoted as a Womersley result, and the fast tail should not be read as a haemodynamic observation. $\lvert\boldsymbol{u}\rvert_{\max}$ in particular is a single staircase-corner voxel and should not be used at all.

13 Performance

Measurements on an Apple M1, Threads backend, four threads, 96^3. Run-to-run spread is under 2%.

These tables were re-measured on D3Q27. The streaming and storage comparison was originally taken on another lattice, one this document no longer covers, so keeping the old numbers would have meant either mislabelling them or deleting the result; it was re-run instead. Both tables come from the same pair of runs (one per precision) and are therefore mutually

Figure 9: Steady inflow at Re $= 50$, speed, after 13 500 steps. Volume rendering: a pale translucent shell wherever the lumen mask has a gradient, diffuse-shaded off that gradient, with speed-coloured interior showing through. Inlet cap green at the aortic root, outlet caps red. A single plane through this vessel cuts it into disconnected islands, which is why the whole volume is cast rather than sliced.



(a) systolic peak, $\phi = 0.5$ (b) diastolic trough, $\phi = 0$

Figure 10: Pulsatile inflow, converged, $\alpha = 8.04$, shared colour scale, with streamlines traced from the inlet cap through the current field. The inset traces (76) with a marker at the current phase. At systole the root drives a visible jet; at the diastolic trough it has gone pale while the descending aorta carries the fastest flow in the vessel — the antiphase fast tail of §12.4, which is a storage-and-release effect and not a Womersley lag. Both panels are from the same beat, at 2.9 viscous diffusion times, where every speed statistic repeats to eight significant figures from beat to beat.

consistent, but they are **2–3× faster than the figures this section carried previously** — the earlier numbers predate the optimisations listed at the end of this section. Rates below are the current build.

streaming and storage (BGK, D3Q27)		bytes/node	MLUPS FP64	MLUPS FP32
two-lattice	raw	432	37.8	46.9
Esoteric Pull	raw	216	**39.5**	**52.9**
Esoteric Pull	shifted	216	39.0	49.5

collision operator (Esoteric Pull + shifted)	BGK	TRT	MRT	central moments
D3Q27, FP64	39.2	35.4	20.7	20.7
D3Q27, FP32	52.0	45.2	21.8	21.6

Caveat. These ratios must be read with care. At ~ 2.3–16.3 GB/s against the ~ 68 GB/s the machine can sustain, this platform is not memory-bound at these rates, so the collision cost shows at full price and the streaming advantage is understated. Lattice Boltzmann is bandwidth-bound on real hardware, and Esoteric Pull moves exactly half the bytes, so its asymptotic gain is $\sim 2\times$. **Re-run `lbm_app` on the target hardware before choosing an operator on cost grounds.**

Optimisations found through the benchmark harness. All were measured rather than assumed:

- Restructuring the streaming hot loop to operate on opposite *pairs* rather than per index removed a runtime `i & 1` test: D3Q27 Esoteric Pull went from 13.5 to 14.0 MLUPS, moving ahead of two-lattice.
- `ProductBasis::pi()` was calling a `constexpr` linear search over the velocity set with *loop-variable* arguments, so it ran live: $2Q^2$ comparisons per node. A precomputed table took the moment operators from 1.8 to 6.6 MLUPS — a 3.7× defect, not a design cost.
- The same class of defect reappeared when the operator was made basis-generic: `p_of(n)` as `n/9`, `(n/3)%3` costs three integer divisions per moment with a runtime index. Table lookups recovered 9%.
- Hoisting the equilibrium out of TRT's non-vectorising pair loop gained 37% on D3Q27.

Compacting the neighbour list to the odd half was also tried, made no measurable difference, and was reverted.

14 Building and running

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release \
      -DKokkos_ENABLE_SERIAL=ON -DKokkos_ENABLE_THREADS=ON
cmake --build build -j8
cd build && ctest --output-on-failure
```

Precision is a configure-time switch, `-DLBM_PRECISION=float` (default `double`). Kokkos is resolved in the order: an installed `find_package` (use this on a cluster, where Kokkos is built once against the correct CUDA architecture), then `extern/kokkos` as a submodule, then `FetchContent`. On a CUDA machine, configure with `-DKokkos_ENABLE_CUDA=ON -DKokkos_ARCH_<ARCH>=ON`.

executable	role
`tests/*`	unit tests (lattice, streaming, equilibrium, collision, moments)
`validation/poiseuille`	walls, magic parameters, resolution behaviour
`validation/decaying_flows`	convergence, isotropy, cross-lattice consistency
`validation/galilean`	frame invariance across operators
`validation/thermal`	scalar transport + one coupled convection case
`validation/mhd`	resistive decay, Alfvén wave, Orszag–Tang divergence
`validation/grid_convergence`	full lattice × operator sweep, CSV
`validation/natural_convection`	Rayleigh sweep against the benchmark
`validation/orszag_tang`	published comparison and stability runs
`lbm_app`	throughput benchmark

The four in the lower block are analysis runs, not regression tests: they take minutes and are excluded from `ctest`.

15 Known limitations

Stated plainly, because each has been measured.

1. **No MPI.** The code is single-rank. `Domain` carries an `mpi_halo` hook and the halo machinery exists, but domain decomposition is not implemented. Note that Esoteric Pull will require a two-way, parity-dependent halo exchange (§3.3).

2. **Convergence studies must watch the Mach number.** The inlet Poiseuille of §11.6 reaches a velocity-dependent error term that does not refine away, and at $U_0 = 0.005$ it dominates by $L_y = 641$ and pulls the fitted rate from 2.067 down to 1.964. This is the cubic aliasing defect $c_{i\zeta}^3 = c_{i\zeta}$ discussed in [12], not a boundary-condition problem: it survives moving the outlet, changing its order, and refining. Any new convergence test should sweep U_0 before a shortfall in the fitted rate is blamed on the scheme.

3. **The scalar module has no shifted-storage default.** T_{ref} must be set to the mean temperature by the user for FP32 accuracy; leaving it at 0 reproduces unshifted storage.

4. **$\nabla\cdot\boldsymbol{b}$ is not preserved to machine precision** (order 1.65 convergence). No divergence cleaning is implemented.

5. **Magnetic walls impose a prescribed $\boldsymbol{b}$ only.** The moment condition of §8.4 sets $\boldsymbol{b}$ to a given value at the wall, which is what Maxwell's equations give for the Hartmann geometry. Genuinely conducting and insulating walls — where the wall's conductivity determines the field rather than the field being prescribed — are not implemented and are not faked.

6. **FP32 cannot resolve the finest convergence tests.** At $H = 64$ the discretisation error falls below the FP32 round-off floor; the affected assertions are FP64-only by design.

7. **Stability results are bounded by the run length.** "Stable through $t = 1.20\,\mathrm{s}$" means no divergence was detected before that cap, not that the run is unconditionally stable.

8. **The regularised condition has a curvature-induced wall slip.** Whenever the tangential velocity has wall-normal curvature the effective wall is displaced by

$$\text{offset} \times W = \frac{4}{3}\,\frac{\omega-1}{\omega^2} \qquad \Longleftrightarrow \qquad \delta u_{\text{slip}} = -\frac{2}{3}\,\frac{\omega-1}{\omega^2}\,\frac{\partial^2 u_\parallel}{\partial n^2}. \tag{78}$$

It is zero for a linear profile — Couette is exact to 4×10^{-15} at every τ from 0.6 to 2.0 — and zero at $\omega = 1$, but it is the whole of the offset column in §10.10 and it limits Ra_c for the on-node wall pair in §10.7. It is intrinsic to truncating $f^{(1)}$ at second Hermite order, which is what "regularised" means: a property of the method, not a defect of this implementation. No correction is applied, but since (57) is exact a user needing better than $O(h)$ wall placement in a curved profile can run at $\omega = 1$ or subtract it.

Derivation. The source is in $f^{(2)}$ — curvature cannot appear in $f^{(1)}$, which carries only first derivatives:

$$f^{(2)} \supset \frac{1-\omega/2}{\omega^2}(\boldsymbol{c}_i\cdot\nabla)^2 f^{\mathrm{eq}} \supset \frac{1-\omega/2}{\omega^2}\frac{w_i}{c_s^2}c_{iy}^2 c_{ix}\,\rho\, u'', \tag{79}$$

odd in $\boldsymbol{c}_i$, hence exactly what the bounce-back scaffold reverses. Its half-space contraction is an exact lattice sum,

$$\sum_{c_{iy}>0} w_i c_{ix}^2 c_{iy}^3 = \frac{1}{18} \quad\Longrightarrow\quad \delta\Pi_{xy} = -\frac{1}{3}\frac{1-\omega/2}{\omega^2}\rho\, u''. \tag{80}$$

Decomposing f^{neq} at an interior node of a converged channel reproduces this source with ratio 1.000 at $\tau = 0.6, 0.8, 1.0, 1.5$, the probe first validated against $\sum_i f^{\mathrm{neq}} = 0$ (to 10^{-17}) and $\sum_i \boldsymbol{c}_i f^{\mathrm{neq}} = -\boldsymbol{F}/2$.

For the wall itself, group the populations by c_{iy} and take the x-odd combination $g_{c_y} = f_{(1,c_y)} - f_{(-1,c_y)}$. The steady recurrences give $(1-\frac{\omega}{2})r^2 - (2-\omega)r + (1-\frac{\omega}{2}) = 0$, i.e. $(r-1)^2 = 0$: a double hydrodynamic root and *no* kinetic boundary layer, so the discrete solution is exactly the parabola and the whole effect sits in the closure. With $u = 3(g_+ + g_-) + C$, $C = F[(2-\omega)/\omega + 3/2]$, the reconstruction giving $G_{c_y} = -w_{c_y}F/c_s^2 + 2w_{c_y}c_y\Pi_{xy}/c_s^4$ and the corrected scaffold giving $\Pi_{xy} = -2\hat{g}_- - F/6$, the wall closes to $g_+(0) + g_-(0) = -F/6$.

The step a one-way argument cannot see is that node 0 collides against the *imposed* $\boldsymbol{u}_w = 0$ rather than the bulk-extrapolated velocity, so continuity with node 1 requires $g_+(0)|_{\mathrm{BC}} = p(0) + \omega U/[6(1-\omega)]$. Eliminating,

$$U\frac{2-\omega}{6(1-\omega)} = \frac{C}{3} - \frac{F}{6}, \qquad 2C - F = \frac{4F}{\omega} \quad\Longrightarrow\quad U = \frac{4F(1-\omega)}{\omega(2-\omega)}, \tag{81}$$

and dividing by $du/dy|_0 = AW$ with $A = 3F\omega/(2-\omega)$ cancels both F and $(2-\omega)$, leaving (57). The coefficient $\frac{4}{3} = 2 \times \frac{2}{3}$ is rational because both factors are exact lattice quantities. Measured: exact at $\tau = 0.6, 0.7, 0.9, 1.0$, within 0.05% at $\tau = 0.8$, 1% at $\tau = 2$ where $\nu = 0.5$ is far outside the hydrodynamic regime assumed.

Note that the force corrections of §8.1 *raise* the reported Poiseuille error at $\tau = 0.8$, from 9.07×10^{-3} to 1.71×10^{-2}, because the force defect and the curvature slip have opposite signs there and were partially cancelling. That cancellation was accidental and τ-specific; at $\tau = 1$ the corrected condition is exact.

9. **ω_3 (bulk) in the MHD central-moment scheme** is not specified in [6]; the value 1 was used here and is exposed as a flag. The stability comparison should be read as close rather than exact reproduction.

10. **No wall compliance.** The arbitrary-geometry path (§12) treats the wall as rigid and static: halfway bounce-back on solid voxels, with no fluid–structure coupling. For the aorta this is the largest single departure from the physics being depicted. A real aorta distends in systole and stores volume elastically, and the pulse wave propagates along the compliant wall at ∼5 m/s; that Windkessel behaviour is the dominant feature of aortic haemodynamics. With a rigid wall the vessel responds essentially instantaneously and there is no pulse wave at all — the pulsatile case of §12.4 produces a pulsatile *flow rate*, not a pulse *wave*, and should not be read as the latter. Nothing in the solver approximates compliance, and none is faked.

11. **Voxelised walls are a staircase, and wall shear stress from them is not trustworthy.** The surface is quantised to half-integer lattice planes. Halfway bounce-back is second order for a wall that genuinely sits at the halfway position, but an oblique vessel wall does not, so the geometric error is $O(h)$ and the effective radius is uncertain by about half a voxel. No interpolated curved-boundary treatment (Bouzidi, Filippova–Hänel) is implemented. Two consequences are measured rather than asserted: the staircase corners produce the speed outlier of §12.4 ($p_{99.9} = 0.026$ against $p_{100} = 0.070$), and because the central-moment operator relaxes all moments of order ≥ 3 to equilibrium there is no free rate left to set the magic parameter — the equivalent $\Lambda = (\tau - \frac{1}{2})/2 = 0.0197$ sits nearly ten times below the 3/16 at which the bounce-back wall is τ-independent (§10.1), so the effective no-slip plane is offset and moves with viscosity. Voxel-scale gradients at the wall should therefore not be used to report WSS.

12. **Performance figures are from a compute-bound machine** and do not predict GPU behaviour (§13).

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